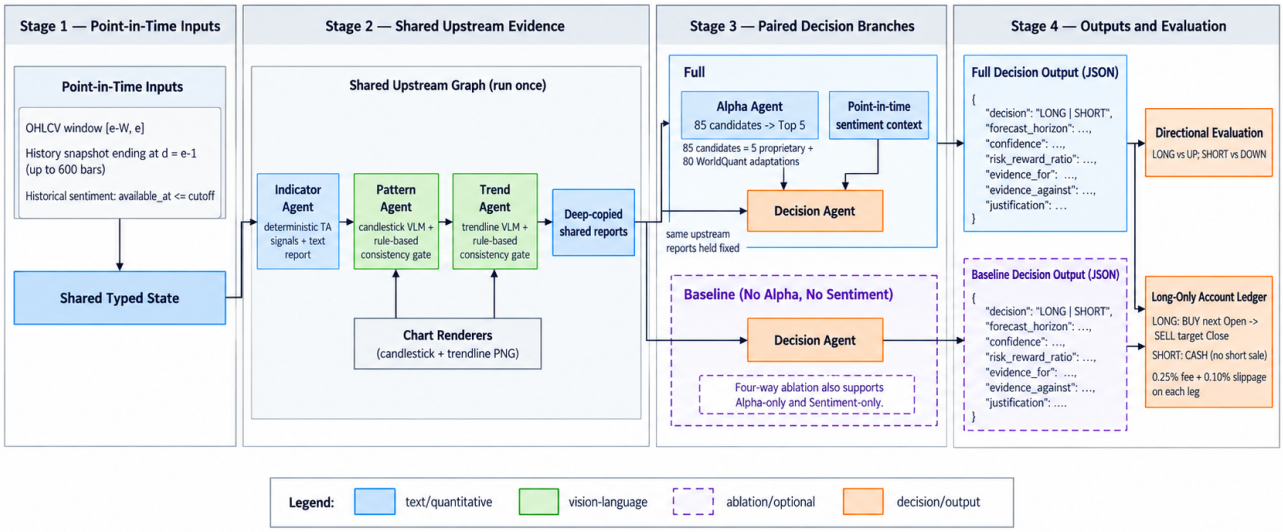
Graphical Abstract

Evaluating Quantitative Evidence in Multi-Agent Decision Support for Vietnamese Equities*

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Highlights

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- A frozen study evaluates 102 Base–Full pairs across six historical blocks.
- CASH exceeds Full in accuracy; price logistic has higher balanced accuracy.
- A separate 270-decision panel tests complete factor-value placebo prompts.
- Full has negative mean net return under explicit execution assumptions.
- The evidence does not establish alpha value or predictive superiority.

Evaluating Quantitative Evidence in Multi-Agent Decision Support for Vietnamese Equities

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Abstract

We evaluate a five-agent decision-support architecture for Vietnamese equities using frozen historical inputs and a separate audit of legacy predictions. The original study contains 102 paired market units across nine equities, six blocks in 2023–2026 and twelve decision dates, with 25 UP and 77 DOWN net-cycle outcomes. Full achieves 62.75% accuracy versus 75.49% for CASH, balanced accuracy of 48.31%, and Matthews correlation of −0.035; price-only logistic regression achieves 55.61% balanced accuracy. Full's mean net return across 51 symbol–block accounts is −0.88%, versus zero for CASH. The descriptive six-block Full–Base balanced-accuracy interval is $[-5.91, 8.00]$ percentage points. The original 474 research decisions and limited repeat panels are archived, excluding ten smoke decisions. Actual-prompt inspection reveals that the original alpha distiller omitted the five factor-value rows. A separately frozen repaired panel therefore executes 270 fresh decisions on 18 preselected units, with complete factor values, factorial branches and a training-donor placebo. Full's balanced accuracy is 44.44%; its mean Full–placebo difference is 6.94 points, but the seven-contrast descriptive envelope is $[-6.55, 20.44]$. All branches have UP recall of 16.67%, so the measured gains concern fewer DOWN errors. These sparse historical panels do not establish general predictive superiority, isolated alpha/sentiment information value or profitability. Unverified served-model revisions and contamination, uncertain news availability and simplified fractional immediate-cash execution limit interpretation. Original compressed, repaired and legacy results remain separate.

Keywords: Multi-agent LLM, Knowledge fusion, Knowledge integration, Financial decision support, Alpha factor selection

1. Introduction

Large language models (LLMs) have recently shown strong potential as reasoning engines in financial analysis. Their value for knowledge-based decision support depends on reliable integration of heterogeneous quantitative, textual, and visual evidence; distributing tasks among agents alone is insufficient. Frameworks such as QuantAgent [1] provide a compelling blueprint for LLM-driven trading, but they primarily address mature, English-language markets and leave the knowledge-integration requirements of emerging economies underexplored.

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Vietnam's equity market presents three challenges that require explicit adaptation. First, a predictive horizon is defined over future bars, whereas an executable cash-equity simulation depends on explicit entry, exit, and position assumptions. Calendar gaps and instrument scope further prevent a bar horizon from being interpreted automatically as an equivalent economic holding period. Second, the dominant financial news source, CafeF, publishes exclusively in Vietnamese, making English-centric sentiment pipelines inapplicable. Third, the shallow cross-sectional universe of HOSE, HNX, and UPCoM calls for alpha factor selection methods suited to single-stock Open, High, Low, Close, and Volume (OHLCV) data instead of the large-universe assumptions of standard factor libraries.

We present a five-agent LangGraph [2] architecture that addresses these gaps through explicit knowledge fusion. An Indicator Agent supplies rule-based technical evidence; an Alpha Agent combines point-in-time selection from 85 OHLCV candidates with separately represented Vietnamese sentiment context; Pattern and Trend agents analyse complementary chart views; and a Decision Agent integrates the resulting reports into a binary LONG/SHORT forecast whose account action is resolved separately. This architecture provides a reproducible basis for integrating heterogeneous quantitative, textual, and visual evidence in an emerging-market decision-support setting. Section 4 documents sentiment construction and provenance controls at the methodological level.

The objective is to test whether the additional quantitative and sentiment bundle improves predictions against appropriate controls. A newly frozen common-snapshot study evaluates 102 Base–Full pairs across six historical blocks, with numerical baselines fitted using purged chronological train/validation data. Full does not exceed CASH in raw accuracy or price-only logistic regression in balanced accuracy, and its mean net account return is negative. Preselected decision-repeat, factorial/placebo and upstream-repeat panels describe variation on limited support. The control removes both factors and sentiment, so the broad Full–Base gap remains a bundle comparison. A separate audit preserves the 180 legacy decisions and their unresolved input/provenance limits; they are not pooled with the new study.

The main contributions are as follows:

- We describe the five-agent implementation and distinguish its information bundle from the Base reports control without claiming isolated component value.
- An independent legacy audit reports class support, confusion matrices, balanced accuracy, MCC and both classes' precision/recall alongside constant baselines.
- A new common-snapshot experiment completes all 474 planned research decisions, including 102 Base–Full pairs and limited preselected repeated panels, and compares train-fitted numerical controls on common support.
- Both-class metrics and separate daily ledgers distinguish CASH, repeated LONG cycles and continuous holding; the negative results do not establish predictive superiority or deployable profitability.

The remainder of this paper is organised as follows. Section 2 reviews related work on LLM-based trading agents, Vietnamese NLP, and quantitative alpha research. Section 3 formalises the problem. Section 4 describes the system architecture. Section 5 presents the experimental setup. Section 6 reports backtest results. Section 7 discusses limitations, and Section 8 concludes.

2. Related works

2.1. LLM Agents in Financial Analysis

The application of LLMs to financial tasks has grown rapidly, spanning earnings call analysis, financial question answering, and report summarisation. FinGPT [3] and BloombergGPT [4] demonstrate that domain-adapted language models outperform general-purpose ones on financial benchmarks. More relevant to our work, TradingAgent [5] and QuantAgent frame trading as a multi-step reasoning problem, delegating sub-tasks-market perception, reflection, and decision-to coordinated LLM agents. Our framework builds directly on this paradigm but extends it with vision-language chart analysis, quantitative alpha integration, and emerging-market adaptations absent from prior systems.

2.2. Technical Analysis and Pattern Recognition

Automating technical analysis has been explored through convolutional neural networks for candlestick pattern recognition [6], reinforcement learning for indicator-based policy learning [7], and hybrid rule-based systems [8]. Recent work has begun replacing handcrafted features with multimodal chart-understanding models. ChartLlama [9] demonstrates broad chart-description capability, but generic chart competence does not establish faithful use of financial-chart evidence. FinChart-Bench finds material instruction-following and spatial-reasoning limitations across 26 vision-language models on financial charts [10]. More specifically, the Martingale Doppelgänger-Eval study shows that observational chart accuracy cannot identify whether a model used local candlestick evidence or merely extrapolated the prevailing trend; matched evidence interventions are required to separate these behaviours [11]. Our two vision agents therefore use deterministic contradiction checks as a safety gate, but we do not interpret that gate as a causal evidence-faithfulness audit.

2.3. Quantitative Alpha Factor Research

Systematic factor construction has a long tradition in quantitative finance, from foundational momentum and value factors [12, 13] to the large-scale formulaic approach of Kakushadze, who codified 101 analytic alpha expressions (WorldQuant 101). Subsequent work adapts formulas to single-asset settings [14] and evaluates factor decay and capacity in different market regimes [15, 16]. Factor mining is nevertheless susceptible to data-snooping and multiplicity: selecting and assessing a model on the same resample can bias its apparent error, whereas an outer validation loop must repeat the complete inner selection procedure [17]. Financial strategy selection adds non-normal returns and the number of trials, which motivate the Deflated Sharpe Ratio [18]. Our executable registry contains five custom formulas and 80 adaptations selected from the 101-formula source catalogue. We dynamically select five candidates per cutoff, but the reported experiment does not use nested validation or a Deflated Sharpe correction; its candidate-level IC and Sharpe values must therefore be read as selection diagnostics rather than confirmatory evidence for individual factors.

2.4. Sentiment Analysis in Finance and Vietnamese NLP

News sentiment has been established as a statistically significant predictor of short-term price movements [19], with transformer-based models such as FinBERT [20] setting the standard for English financial text. For Vietnamese, ViSoBERT provides a BERT-based model pre-trained on large-scale Vietnamese corpora and achieves strong results on Vietnamese sentiment classification benchmarks. Financial-domain validity does not follow automatically from general-language benchmark performance. We therefore use the public `5CD-AI/Vietnamese-Sentiment-visobert` classifier through a local Transformers backend by default, with an optional Hugging Face Dedicated Inference

Endpoint and an explicit lexicon fallback. The actual scoring path is recorded for every newly scored article. Target and related-company sentiment are presented in a separate LLM text context and do not enter the OHLCV factor formulas or ranking. Legacy cache entries that lack scorer metadata are treated as unknown rather than retrospectively attributed to ViSoBERT.

2.5. Trading Systems for Emerging Markets

Research on algorithmic trading in emerging markets is comparatively sparse. Studies on the Vietnamese market have primarily examined market efficiency [21], herding behaviour [22], and the predictive power of classical technical indicators under the price-limit regime [23], but none employs LLM-based agents. Our work develops a multi-agent knowledge-fusion and decision-support framework that records Vietnamese cash-equity execution and settlement assumptions separately from the domestic news ecosystem and single-stock OHLCV data characteristics. Consequently, the model horizon remains an explicit experimental choice instead of being inferred from a universal settlement rule.

2.6. Evaluation Standards and Agentic Benchmarks

Recent literature highlights the critical need for rigorous, standardized evaluation in LLM-driven trading. For instance, FinVision [24] shares our modular multi-agent approach and vision-based technical analysis but places much stronger emphasis on extensive ablations (e.g., of reflection modules) and comprehensive risk reporting (Sharpe, drawdowns). Furthermore, an audit-oriented survey on financial LLMs [25] identifies widespread methodological gaps in time-consistent splits, transaction-cost modeling, and reproducibility. While our framework addresses costs and lookahead leakage, we recognize that our reporting detail currently falls short of the rigorous bar for full reproducibility. Finally, live/prediction-market benchmarks [26, 27] and forward-looking evaluation paradigms like FutureX [28] underscore that static walk-forward backtests–while useful–must ultimately be complemented by live trials and execution realism to substantiate strong claims about agentic efficacy.

3. Problem Formulation

3.1. Market Model and Data

Let $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$ denote the universe of Vietnamese equities listed on HOSE, HNX, or UPCoM. For each asset $s \in \mathcal{S}$, the market is observed as a discrete-time OHLCV series:

$$\mathbf{x}_u = \big(O_u,\, H_u,\, L_u,\, C_u,\, V_u\big) \;\in\; \mathbb{R}_+^5, \quad u = 0, 1, \ldots, T-1, \tag{1}$$

where O_u, H_u, L_u, C_u, V_u denote the open, high, low, closing price, and trading volume at zero-based bar position u respectively. When volume observations are missing for more than 50% of bars, the intra-bar range $H_u - L_u$ is substituted as a liquidity proxy.

We use zero-based row positions and an exclusive window-end boundary. For a test point with boundary e, the analysis window of size W is exactly:

$$\mathbf{X}_e = \big(\mathbf{x}_{e-W},\, \mathbf{x}_{e-W+1},\, \ldots,\, \mathbf{x}_{e-1}\big) \;\in\; \mathbb{R}^{W\times 5}. \qquad (\text{equivalently, } \mathbf{X}_e = \mathbf{X}_{[e-W,e)}) \tag{2}$$

Thus e is not itself a visible bar: the final visible (decision) bar is $d = e - 1$. The next bar, e, is reserved for the earliest simulated entry, and the immutable label target is $h = e-1+L$. A candidate economic exit may use that close only if the separate execution contract accepts it. This mapping is used in the engine, alpha labels, persisted provenance, and the experiments:

$$\boxed{\text{window} = [e-W, e), \qquad d = e-1, \qquad \text{entry} = e, \qquad h = e-1+L.} \tag{3}$$

Algorithm 1: Walk-Forward Evaluation

Input: OHLCV series $\mathbf{X}$, alpha registry $\mathcal{A}$, historical sentiment cache $\mathcal{D}$, window size W, lookahead L, step size S

Output: Directional predictions $\hat{y}_{e,L}$, accuracy, Full–Base difference, and compounded account return

Initialize results list $\mathcal{R} \leftarrow \emptyset$

foreach *test point e in the walk-forward schedule* **do**

- Extract the exact analysis window $\mathbf{X}_e = \mathbf{X}_{[e-W,e)}$
- Compute technical indicators using TA-Lib
- Evaluate all candidate alphas $\alpha^{(k)} \in \mathcal{A}$
- Remove candidates with no fitted transform, fewer than 20 finite evaluation observations, or non-finite ranking metrics to form $\mathcal{V}$
- Min–max normalise each metric over $\mathcal{V}$ and rank using:

$$S_k = 0.40\mathcal{N}_{\mathcal{V}}(|IC_k|) + 0.35\mathcal{N}_{\mathcal{V}}(Acc_k) + 0.25\mathcal{N}_{\mathcal{V}}(Sharpe_k)$$

- Select top-5 alpha factors $\mathcal{A}^*$
- Retrieve historical CafeF sentiment at the decision cutoff $d = e - 1$
- Generate candlestick and trendline chart images from the same exact $\mathbf{X}_e$ window
- Run the Indicator, Alpha, Pattern, and Trend Agents
- Pass all reports to Decision Agent
- Obtain binary prediction $\hat{y}_{e,L} \in \{\texttt{LONG}, \texttt{SHORT}\}$
- Compare prediction with realised return
- Store metrics in $\mathcal{R}$

Compute directional accuracy and Alpha Lift

Compute compounded account return under Eq. (34)

return $\mathcal{R}$

The mapping is positional rather than calendar arithmetic; timestamps are carried alongside every index, so a weekend or holiday gap does not create a synthetic bar. The first complete window is $e = W$, and it is evaluable only when the target is present, i.e., $T \geq W + L$. For $L = 1$, the target is the entry bar ($h = e$), while for $L = 3$ it is two positions after entry ($h = e + 2$). Any boundary whose target is outside the available frame is rejected rather than shortened.

3.2. Directional Prediction Task

The core task predicts whether a candidate LONG cycle is profitable after the configured transaction costs. With $d = e - 1$ and $h = e - 1 + L$, define

$$r^{\text{net}}_{e,L} = \frac{C_h}{O_e}\frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1, \qquad y_{e,L} = \begin{cases} +1, & r^{\text{net}}_{e,L} > 0, \\ -1, & r^{\text{net}}_{e,L} \leq 0. \end{cases} \tag{4}$$

Here O_e is the next-bar entry open and C_h is the target close. The same net Open-to-Close return defines the directional label, the alpha-selection target, and the realised return of an executed LONG. Thus an opening gap and both transaction legs are represented consistently. The reported study uses daily bars with $L = 3$; intraday execution is outside its empirical scope.

A trading system $\mathcal{F}$ maps each analysis window to a binary decision:

$$\hat{y}_{e,L} \;=\; \mathcal{F}\big(\mathbf{X}_e\big) \;\in\; \{-1, +1\}, \tag{5}$$

where $+1$ corresponds to a `LONG` signal and -1 to `SHORT`.

For the reanalysis, the net economic target above is retained unchanged. A separate secondary target is $y^{\text{gross}} = +1$ if $C_{e-1+L}/O_e - 1 > 0$, and -1 otherwise. Gross-target metrics, where reported in the reanalysis package, use the same predictions and are not used to retune models or replace the net target. Thus the reported task is profitable-cycle classification, rather than an unqualified forecast of the sign of the close-to-close price change.

3.3. Alpha Factor Formalisation and Adaptation

An *alpha factor* $\alpha^{(k)}$ is a measurable function that maps a feature frame derived from $\mathbf{X}_e$ to a scalar signal:

$$\alpha^{(k)} : \mathbb{R}^{T_{\text{hist}} \times d} \longrightarrow \mathbb{R}, \tag{6}$$

where d is the number of engineered features and T_{hist} is the historical window used for computation.

The executable registry contains 80 adaptations drawn from the WorldQuant-101 catalogue, not all 101 source expressions, plus five custom formulas. The source expressions were designed primarily for cross-sectional ranking across a broad, liquid equity universe. To apply the registered subset to a single Vietnamese stock, we use a *semantic translation layer* that replaces cross-sectional operators with rolling time-series surrogates. This is an engineering heuristic rather than an invariance-preserving mathematical transformation.

Operator Localisation.. The primary transformation involves the cross-sectional rank(x) operator, which denotes the relative percentile of a stock against its peers. A cross-sectional rank is localised to a 20-bar time-series percentile rank unless the source expression already specifies a time-series rank with its own window. Let $\mathcal{H}_{u,w}$ contain the available preceding observations in that rolling window and let $n_{u,w} = |\mathcal{H}_{u,w}| + 1$. The implementation computes

$$\text{ts_rank}(x_u, w) \;=\; \frac{1}{\max(n_{u,w} - 1, 1)} \sum_{x_v \in \mathcal{H}_{u,w}} \mathbf{1}[x_u > x_v]. \tag{7}$$

Thus ties are not counted as strictly lower values, and an initial partial window is permitted once the operator-specific minimum is met. Other explicit time-series lookbacks are retained from each registered expression and can be substantially longer than 20 bars. Average Daily Volume (ADV_n) is the asset's own n-day rolling mean volume, and VWAP is approximated by the daily typical price $(H_u + L_u + C_u)/3$. Functions $\delta(x, d)$ and $\text{delay}(x, d)$ are implemented as d-period differences and lags. These substitutions remove peer-relative information, so neither the economic interpretation nor the stochastic properties of the source cross-sectional alpha are presumed to survive unchanged.

Consistent Normalisation.. To ensure comparability across candidates with different scales, each adapted raw alpha value is normalised to $[-1, 1]$ via:

$$\tilde{\alpha}_{u,e}^{(k)} \;=\; \tanh\left(\frac{\alpha_u^{(k)} - \mu_{k,e}}{\sigma_{k,e} + \varepsilon}\right), \tag{8}$$

where $\mu_{k,e}$ and $\sigma_{k,e}$ are computed over the finite values in the cutoff-safe calibration snapshot available at decision boundary e, and $\varepsilon > 0$ prevents division by zero. They are recomputed at the next cutoff; they are not estimated from the entry or target bars. The resulting signal $\tilde{\alpha}_u^{(k)} \in (-1, 1)$ is treated as a continuous directional score: positive values favour `LONG` and negative values favour `SHORT`.

3.4. Walk-Forward Alpha Selection

Let $\mathcal{A} = \{\alpha^{(1)}, \alpha^{(2)}, \ldots, \alpha^{(M)}\}$ denote the registry of $M = 85$ candidate alpha functions. Each point-in-time history ends at the decision cutoff and is therefore disjoint from the current entry and target bars. Given the net Open-to-Close execution-return series $r_u^{\text{net},(L)}$ defined as in Eq. (4), the following diagnostics are computed on finite historical observations:

The same matured calibration sample is used to estimate IC sign, orient the signal, calculate the three ranking diagnostics, and select the top five. Consequently, point-in-time truncation prevents future-data leakage but does not provide an independent inner/outer assessment of the adaptive selection rule. The equations below define the implemented selector; they should not be read as post-selection-unbiased estimates of an individual factor's skill.

Information Coefficient (IC).. Following Grinold and Kahn [29], the Spearman rank correlation between the normalised alpha signal and forward returns over the evaluation suffix:

$$\text{IC}_k \;=\; \text{SpearmanCorr}\big(\tilde{\alpha}^{(k)}_{\mathcal{T}_{\text{eval}}},\; r^{\text{net},(L)}_{\mathcal{T}_{\text{eval}}}\big). \tag{9}$$

Directional Accuracy.. The fraction of bars where the predicted direction agrees with the realised direction, restricted to bars with a strong enough signal:

$$\text{Acc}_k \;=\; \frac{1}{|\mathcal{T}^*|} \sum_{u \in \mathcal{T}^*} \mathbf{1}\big[\text{sign}\big(\tilde{\alpha}^{(k)}_u\big) = y^{(L)}_u\big]\,, \tag{10}$$

where $\mathcal{T}^* = \{u : |\tilde{\alpha}^{(k)}_u| > 0.05\}$ excludes near-zero signals and $y^{(L)}_u$ is the canonical binary label from Eq. (4); hence a zero net return is `DOWN`, not a third class.

Long-Only Win Rate.. The precision of the alpha on bullish signals:

$$\text{LongAcc}_k \;=\; \frac{\sum_u \mathbf{1}\Big[\tilde{\alpha}^{(k)}_u > 0.05 \;\wedge\; r^{\text{net},(L)}_u > 0\Big]}{\sum_u \mathbf{1}\Big[\tilde{\alpha}^{(k)}_u > 0.05\Big]}. \tag{11}$$

Horizon Sharpe diagnostic.. For ranking, define $z^{(k)}_u = \text{sign}(\tilde{\alpha}^{(k)}_u) r^{\text{net},(L)}_u$ on the finite historical sample. Following Sharpe [30], the diagnostic is:

$$\text{Sharpe}_k \;=\; \sqrt{\frac{252}{L}} \cdot \frac{\mathbb{E}_{u \in \mathcal{T}_{\text{eval}}}\Big[z^{(k)}_u\Big]}{\text{Std}_{u \in \mathcal{T}_{\text{eval}}}\Big[z^{(k)}_u\Big] + \varepsilon}. \tag{12}$$

Sortino, maximum drawdown, hit rate, long-only accuracy, and average trade are retained as descriptive candidate diagnostics but do not enter the composite selection score.

Candidates with no fitted transform, fewer than 20 finite evaluation observations, or a non-finite value for any ranking metric are excluded before scaling. Let $\mathcal{V} \subseteq \mathcal{A}$ be the remaining eligible set and let $q \in \{|\text{IC}|, \text{Acc}, \text{Sharpe}\}$. The implementation applies candidate-wise min–max scaling:

$$\mathcal{N}_{\mathcal{V}}(q_k) = \begin{cases} \dfrac{q_k - q_{\min}}{q_{\max} - q_{\min}}, & q_{\max} - q_{\min} \geq \varepsilon_{\text{rank}}, \\ \dfrac{1}{2}, & \text{otherwise}, \end{cases} \tag{13}$$

where $q_{\min} = \min_{j \in \mathcal{V}} q_j$, $q_{\max} = \max_{j \in \mathcal{V}} q_j$, and $\varepsilon_{\text{rank}} = 10^{-9}$. The default composite selection score is therefore:

$$S_k \;=\; 0.40\,\mathcal{N}_{\mathcal{V}}(|\mathrm{IC}_k|) \;+\; 0.35\,\mathcal{N}_{\mathcal{V}}(\mathrm{Acc}_k) \;+\; 0.25\,\mathcal{N}_{\mathcal{V}}(\mathrm{Sharpe}_k). \tag{14}$$

The weights prioritise association and directional accuracy while retaining a risk-adjusted return diagnostic. Long-only accuracy is reported but excluded from ranking to avoid favouring factors solely because the sample trends up.

Signal orientation is inferred from the sign of IC on the cutoff-safe historical snapshot. A negative-IC candidate is multiplied by -1 for its historical directional metrics and for the current inference value. All labels used for this operation have matured by the decision cutoff, but estimating orientation and ranking on the same finite history remains a selection risk discussed in Section 7.

Let $K = \min(5, |\mathcal{V}|)$. The top candidates are selected as:

$$\mathcal{A}^* \;=\; \operatorname*{arg\,max}_{\mathcal{B} \subseteq \mathcal{V},\, |\mathcal{B}| = K} \sum_{\alpha^{(k)} \in \mathcal{B}} S_k. \tag{15}$$

3.5. Sentiment Signal Construction

Let $\mathcal{D}_e = \{d_1, d_2, \ldots, d_{|\mathcal{D}_e|}\}$ be the set of CafeF articles available to the system for asset s at the timestamp τ_d of decision bar $d = e - 1$. Each article is assigned a score by the scorer actually recorded for that article:

$$\rho_i \;=\; \mathrm{Scorer}_i(d_i) \;\in\; [-1,\, 1], \tag{16}$$

where $\rho_i = +1$ denotes strongly positive, -1 strongly negative, and 0 neutral sentiment. For newly scored records, Scorer_i is either `5CD-AI/Vietnamese-Sentiment-visobert`, executed locally through Transformers or through a configured Hugging Face Dedicated Inference Endpoint, or the deterministic lexicon fallback. In automatic mode, a configured dedicated endpoint is attempted first and local inference is used if it fails; without an endpoint URL, local inference is selected directly. The classifier's official label mapping is $0 \mapsto$ NEG, $1 \mapsto$ POS, and $2 \mapsto$ NEU. Each record stores `scorer`, `scorer_backend`, and `is_fallback`. These fields identify the effective execution path but do not constitute an immutable model-revision digest. Legacy scores without these fields are marked `legacy-unknown`; no model identity is inferred from their numeric values.

The unweighted mean sentiment score for descriptive reporting is:

$$\bar{\rho}_e \;=\; \frac{1}{|\mathcal{D}_e|} \sum_{i=1}^{|\mathcal{D}_e|} \rho_i. \tag{17}$$

For the causal sentiment context, each dated article receives an exponential weight based on its age at cutoff τ_d:

$$\Delta_i(\tau_d) = \tau_d - \mathrm{date}(d_i), \tag{18}$$

$$w_i(\tau_d) = 2^{-\Delta_i(\tau_d)/h}, \qquad h = 30 \text{ days}, \tag{19}$$

$$S_{\text{decay},e} = \frac{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)\rho_i}{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)}, \tag{20}$$

$$\mathrm{Rel}_{\text{sent},e} \;=\; S_{\text{decay},e} \;-\; \frac{1}{|\mathcal{R}_e|} \sum_{s' \in \mathcal{R}_e} S^{(s')}_{\text{decay},e}, \tag{21}$$

where $\mathcal{R}_e$ contains only related companies with at least three valid dated-and-scored articles at cutoff τ_d. The exponential mean makes recent articles more influential without comparing a batch mean against the same batch from which it was calculated. $\mathrm{Rel}_{\text{sent},e}$ compares like-for-like time-decayed scores.

Batch definition and low-news stability.. The set $\mathcal{D}_e$ contains all dated articles for target stock s in the 90-day sliding window ending at cutoff τ_d. The complete compact observation set is used for Eq. 20; the 15-article list stored for display is not the normalization sample. With fewer than three valid dated-and-scored observations, $S_{\text{decay},e}$ is marked unavailable rather than assigned a neutral zero. Likewise, $\text{Rel}_{\text{sent},e}$ is unavailable when $\mathcal{R}_e$ is empty. The effective sample size, invalid-date/score counts, and number of post-cutoff observations excluded are stored in run provenance. Neither context variable enters any numerical alpha formula or the registry ranking.

To prevent look-ahead leakage during backtesting, every cached article stores timezone-aware `published_at` and `available_at` fields in `Asia/Ho_Chi_Minh`. The cache is keyed by (s, τ_d) and returns only articles satisfying $\text{available_at}(d_i) \leq \tau_d$. CafeF records having date-only precision are assigned a conservative availability time of 00:00 on the following local day; records without a parseable date are rejected in research mode and never used to fill a sparse historical window. All 180 articles in the nine supplied cache artifacts contain parseable dates, so the previous undated fallback was a latent risk rather than evidence that those reported runs actually consumed undated articles.

3.6. Multi-Agent Decision Function

The full system decomposes $\mathcal{F}$ into five sub-functions corresponding to the five specialised agents:

$$\mathbf{r}^{\text{ind}} = \mathcal{F}_{\text{ind}}\big(\mathbf{X}_e\big), \tag{22}$$

$$\mathbf{a}_e = \mathcal{F}_{\text{factor}}\big(\mathbf{X}_e,\ \mathcal{A}^*\big), \tag{23}$$

$$\mathbf{c}_e^{\text{sent}} = \mathcal{F}_{\text{context}}\big(\mathcal{D}_e,\ S_{\text{decay},e},\ \text{Rel}_{\text{sent},e}\big), \tag{24}$$

$$\mathbf{r}^{\text{alpha}} = \mathcal{F}_{\text{text}}\big(\mathbf{a}_e,\ \mathbf{c}_e^{\text{sent}}\big), \tag{25}$$

$$\mathbf{r}^{\text{pat}} = \mathcal{F}_{\text{pat}}\big(\mathbf{I}_e^{\text{kline}}\big), \tag{26}$$

$$\mathbf{r}^{\text{trend}} = \mathcal{F}_{\text{trend}}\big(\mathbf{I}_e^{\text{trend}}\big), \tag{27}$$

where $\mathbf{I}_e^{\text{kline}}$ and $\mathbf{I}_e^{\text{trend}}$ are the candlestick and trendline-annotated chart images rendered from $\mathbf{X}_e$, and $\mathbf{c}_e^{\text{sent}}$ is a separate textual context containing the target and related-company sentiment summaries. The factor values in $\mathbf{a}_e$ are functions of $\mathbf{X}_e$ only; the LLM compares them qualitatively with $\mathbf{c}_e^{\text{sent}}$ when producing $\mathbf{r}^{\text{alpha}}$.

The Decision Agent aggregates all four reports through structured chain-of-thought prompting [31] and produces the final binary signal:

$$\hat{y}_{e,L} = \mathcal{F}_{\text{dec}}\big(\mathbf{r}^{\text{ind}},\ \mathbf{r}^{\text{alpha}},\ \mathbf{r}^{\text{pat}},\ \mathbf{r}^{\text{trend}}\big) \in \{-1, +1\}. \tag{28}$$

The forced binary output constraint ensures $\hat{y}_{e,L} \neq 0$ for every valid *prediction*; it does not imply an always-invested account. Under the default long-only action map, `SHORT` becomes `CASH`. `LONG` executes a complete next-open-to-target-close cycle when valid positive prices and timestamps are available. This separation permits binary directional evaluation without misrepresenting a bearish prediction as an executed short sale.

3.7. Walk-Forward Evaluation Protocol

We adopt a rolling walk-forward evaluation protocol commonly used in quantitative trading system evaluation [32].

Given a historical series of T bars, valid test points are exclusive boundaries with $W \leq e \leq T - L$. The implementation uses the latest available boundaries at spacing S (up to N_{test} points) and processes them in chronological order:

$$\mathcal{T} = \big\{\, e_i = T - L - (i-1)\cdot S \ \mid\ i = 1, 2, \ldots, N_{\text{test}} \,\big\}, \tag{29}$$

retaining only values $e_i \geq W$ before reversing the list for execution. Thus $e = W$ is included whenever it is among the requested points, and every selected point satisfies the complete-window and complete-target conditions. Here S is the step size between consecutive test points and $N_{\text{test}} = 20$ in the legacy experiments. The new common-snapshot study instead uses the explicit block schedule in Section 5.6; it does not treat 102 cross-symbol units as 102 independent dates.

The three primary evaluation metrics are:

$$\text{Acc} = \frac{1}{|\mathcal{T}|} \sum_{e \in \mathcal{T}} \mathbf{1}\big[\hat{y}_{e,L} = y_{e,L}\big], \tag{30}$$

$$\Delta\text{Acc} = \text{Acc}_{\text{Full}} - \text{Acc}_{\text{No-}\alpha}, \tag{31}$$

$$q_e = a_e \left[\frac{C_{e-1+L}}{O_e} \frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1 \right], \tag{32}$$

$$W_{k+1} = W_k \left(1 + q_{e_k}\right), \qquad W_0 = 1, \tag{33}$$

$$R_{\text{acct}} = \frac{W_K}{W_0} - 1, \tag{34}$$

where ΔAcc is the Full–Base raw-accuracy difference, called *Alpha Lift* only in legacy artifacts. Because Base removes both factors and sentiment, this difference cannot identify an isolated alpha effect. In the economic simulation, $a_e = 1$ for `LONG` and $a_e = 0$ for `SHORT`. Each LONG allocates all available cash at O_e, pays fee and slippage on the BUY leg, liquidates all shares at C_{e-1+L}, and pays the same rates on the SELL leg. The configured values are $c_{\text{fee}} = 0.25\%$ and $c_{\text{slip}} = 0.10\%$ per leg. Each test point is a closed cycle and the implementation requires $S \geq L$, so no shares carry into the next point. SHORT forecasts hold cash and incur neither fee nor slippage. Let $\rho_k = W_{k+1}/W_k - 1 = q_{e_k}$ denote the cycle return, including zeros for CASH. The account risk metrics are

$$\text{Sharpe}_{\text{acct}} = \sqrt{252}\, \frac{\mathbb{E}[\rho_k]}{\text{Std}[\rho_k] + \varepsilon}, \tag{35}$$

$$\text{MDD}_{\text{acct}} = \max_{0 \leq k \leq K} \left(1 - \frac{W_k}{\max_{0 \leq i \leq k} W_i} \right). \tag{36}$$

The Sharpe risk-free rate is zero, and the population standard deviation is used. Maximum drawdown is a relative wealth loss and includes the initial wealth anchor W_0, so an initial losing cycle is not omitted.

Finally, let $\mathcal{X} = \{e_k : a_{e_k} = 1\}$ be the executed-position sample. Trade-level summaries are

$$\text{HitRate} = \frac{1}{|\mathcal{X}|} \sum_{e \in \mathcal{X}} \mathbf{1}[q_e > 0], \qquad \text{AvgTrade} = \frac{1}{|\mathcal{X}|} \sum_{e \in \mathcal{X}} q_e, \tag{37}$$

with both defined as zero when $\mathcal{X}$ is empty. SHORT/CASH cycles do not dilute either executed-trade denominator. Risk ratios are reported as zero when their dispersion denominator does not exceed the numerical tolerance. The cost rates are research allowances rather than a claim about a particular venue's realised fee schedule.

Report interventions and interaction. Let B, A, S, AS denote the absence/presence of the alpha and sentiment report sections, respectively, with shared upstream reports. For a metric M, the bundle

contrast is $M(AS) - M(B)$; the alpha-report contrasts are $M(A) - M(B)$ without sentiment and $M(AS) - M(S)$ with sentiment. The sentiment-report contrasts are $M(S) - M(B)$ and $M(AS) - M(A)$. The factorial interaction is

$$I_M = M(AS) - M(A) - M(S) + M(B). \tag{38}$$

These are interventions on implemented reports, not automatically isolated information effects: report structure, compressed content, rules and hosted decision variation can change the response. Information attribution requires a valid matched-content control and actual prompt inspection. A positive gap over the better single component is a different, nonlinear statistic.

4. System architecture

4.1. Overview

The implemented runtime is a LangGraph directed acyclic graph with five specialised agents and a shared typed state object. It acts as an explicit knowledge-integration layer over technical, quantitative, textual, and visual evidence. Its default configuration names `openai/gpt-oss-20b` for text and `qwen/qwen3.8-27b` for vision through Groq. The backtest path enforces point-in-time input windows, shared upstream reports for paired variants, structured binary decisions, and a separate compounded-account ledger.

4.2. Indicator Agent

The Indicator Agent computes five momentum/trend indicators using TA-Lib over the observed OHLCV window: MACD [33], RSI [34], Stochastic Oscillator [35], Williams %R [36], and ROC [37]. Table 1 summarises the ordered rules; Appendix Appendix E gives equality, fallback and missing-data conventions. Classification and counts are computed in Python. The backtest sends a deterministic report without an Indicator LLM call; in live mode, the interpretive prompt instructs the LLM to copy the supplied signals/counts. This instruction is not a guarantee of model compliance. A directional plurality determines consensus; confidence is High for at least four agreeing signals, Medium for three, and Low otherwise. Equal bullish/bearish counts yield MIXED.

Bollinger Bands, ADX and directional-index arrays are declared state slots, not additional votes. The current Indicator node invokes only the five tools above and returns `indicator_report` plus `messages`; it does not populate even the declared five-indicator numeric arrays. Bollinger features are computed independently in the Alpha path and may enter selected alpha evidence. ADX is not called or forwarded by the Indicator node. A declared schema field alone does not establish that the Decision Agent received that feature.

4.3. Alpha Agent

The Alpha Agent serves two explicitly separated functions: OHLCV-only quantitative factor evaluation and qualitative comparison with Vietnamese sentiment supplied as textual LLM context.

Alpha selection.. An offline registry of exactly 85 candidate alpha functions is maintained, comprising five custom single-stock momentum and reversion formulas (Table 2) and 80 adaptations drawn from the WorldQuant 101 Formulaic Alphas. Of the original 101 WorldQuant expressions, only 80 are retained because the remaining 21 cannot be translated into a meaningful single-stock OHLCV setting without introducing information that is unavailable in our pipeline or fundamentally altering the original operator semantics. In particular, these excluded expressions rely on cross-sectional

Table 1: Technical indicators and trading signal conditions

Indicator	**Parameters**	**Bullish condition**	**Bearish condition**
MACD	12, 26, 9	Cross up first; otherwise positive, nondecreasing histogram	Cross down first; otherwise negative, nonincreasing histogram
RSI	period = 14	$r \leq 30$ or $55 \leq r < 70$; in $45 < r < 55$, rising three-value trend	$r \geq 70$ or $30 < r \leq 45$; in $45 < r < 55$, falling three-value trend
Stochastic	14, 3, 3	Cross up first; otherwise $K \leq 20$, then middle-zone position rule	Cross down first; otherwise $K \geq 80$, then middle-zone position rule
Williams %R	period = 14	$w \leq -80$; otherwise $-80 < w < -20$ with rising three-value trend	$w \geq -20$; otherwise $-80 < w < -20$ with falling three-value trend
ROC	period = 10	ROC $> +0.5\%$	ROC $< -0.5\%$

ranking or neutralisation across multiple contemporaneous assets, industry or market-group information, or other inputs/operators that do not admit a defensible time-series-only surrogate under our single-symbol formulation. We therefore exclude them rather than force an arbitrary approximation. Consequently, 101 denotes the size of the original source catalogue, whereas 80 denotes the subset considered adaptable to our single-stock setting, and 85 denotes the final executable candidate registry after adding five custom factors.

For the retained expressions, cross-sectional operators are replaced by single-asset time-series surrogates as formalised in Eq. (7). This translation is heuristic: without contemporaneous peer dispersion, a rolling percentile answers where is this stock relative to its own recent history?" rather than where is this stock relative to other stocks now?" The adaptation can therefore change factor scale, persistence, turnover, and economic meaning. The present experiments do not estimate per-factor IC-decay curves, lagged signal autocorrelation, or turnover profiles, and do not claim that the adapted factors inherit the properties of their WorldQuant sources. When the volume series is unavailable or sparse, intra-bar price range is used as a proxy, creating an additional interpretation change for volume-dependent expressions.

For a given symbol and timeframe, the available point-in-time history is truncated to at most 600 bars ending at the decision cutoff. Features (returns, log-volume, VWAP proxy, and ADV20) are then constructed. Candidate metrics use only matured net Open-to-Close labels within this snapshot; the final L rows are excluded because their targets are unavailable at the cutoff. A zero net return is assigned `DOWN`, consistent with the requirement that a LONG must remain profitable after costs.

The selector computes Spearman Information Coefficient (IC), directional accuracy on signals above the configured absolute threshold, and a horizon Sharpe diagnostic. Long-only win rate and other economic summaries are retained for diagnosis but do not enter the composite score.

A candidate is eligible only when it has at least 20 finite observations and finite ranking metrics. Ineligible candidates are removed *before* the per-metric min–max bounds are computed. Using $\mathcal{N}_{\mathcal{V}}$ from Equation (13), the default composite score is:

$$S = 0.40\mathcal{N}_{\mathcal{V}}(|IC|) + 0.35\mathcal{N}_{\mathcal{V}}(Acc) + 0.25\mathcal{N}_{\mathcal{V}}(Sharpe) \tag{39}$$

If a metric is constant across the eligible set, its normalised value is 0.5 for every candidate. The

displayed coefficients are the default runtime weights; robustness configurations replace them with recorded alternatives. Long-only accuracy is excluded from ranking to avoid rewarding a factor solely because the historical sample trends upward.

Up to five eligible candidates are selected for the current prediction. When historical IC is negative, the signal is multiplied by -1 for metric computation and current inference. All observations used for this orientation precede the decision cutoff, which prevents lookahead into the evaluated trade. It does not eliminate selection bias: the same trailing observations determine the IC sign, score all 85 candidates, and choose the five winners. A noisy negative IC can therefore be converted into an apparently useful positive orientation, and the maximum over many noisy scores is optimistic even when every timestamp is cutoff-safe. No nested inner-selection/outer-evaluation loop, Deflated Sharpe Ratio, or rolling inclusion/sign-stability diagnostic is applied in the reported study. Candidate IC and Sharpe values are consequently descriptive ranking statistics, not multiplicity-adjusted evidence that a particular factor is genuinely profitable. Nested validation that repeats orientation and top-five selection inside every training fold [17], together with Deflated Sharpe or false-discovery control [18, 15], is required for stronger factor-level claims.

Table 2: Five custom alpha formulas

Alpha	**Name**	**Formula**
α_1	Flow-Driven Momentum	$\tanh(\mathrm{ROC}(1)_{\mathrm{adj}} \times \mathrm{VolSurge} \times 0.5)$
α_2	MACD-Flow Proxy	$\tanh(\mathrm{MACDHist}_{\mathrm{norm}} + \mathrm{ROC}(1)_{\mathrm{adj}} \times 0.5)$
α_3	Liquidity Void Reversion	$\tanh\big(-0.8 \times \mathrm{ZScore}\big(\frac{\mathrm{Close}-\mathrm{SMA}_5}{\mathrm{Close}}\big)\big)$
α_4	Bollinger Squeeze & Flow	$\tanh((\%B - 0.5) \times \mathrm{VolZ}_5 \times 1.5)$
α_5	Order Flow Exhaustion	$\tanh(\mathrm{ROC}(1)_{\mathrm{adj}} \times \mathrm{ClosePos} \times 3.0)$

Custom-factor interpretation.. The five custom formulas are engineering heuristics, not evidence of ownership or independently validated novelty. Names such as Flow-Driven Momentum and Order Flow Exhaustion are retained identifiers. Here "flow", "liquidity" and "buying pressure" denote OHLCV price/volume proxies; there are no signed-trade, aggressor-side, order-book, queue or institutional-flow observations in these formulas. Intra-bar range substitution, when used, is a price proxy rather than measured share volume.

For α_4, VolZ_5 is signed. A positive value preserves the sign of $\%B - 0.5$, a negative value reverses it, and zero suppresses it. Thus $|\mathrm{VolZ}_5| > 0$ does not establish volume expansion. The formula tests neither a temporal midline crossing nor a band-width squeeze. Appendix Appendix D gives signed examples and the unchanged prototype thresholds. Legacy factor names and breakout/volume-confirmation wording are retained in runtime/prompt archives for replication; they must not be read as a validated mechanism. A positive-volume gate such as $\max(\mathrm{VolZ}_5, 0)$ would be a different, separately registered experiment and is not substituted into the reported results.

Sentiment integration.. Articles from CafeF are crawled for the target ticker and up to ten related companies identified by the LLM. The scorer supports four explicit backend policies: `auto`, `local`, `endpoint`, and `lexicon`. The default `auto` policy calls a user-provisioned Hugging Face Dedicated Inference Endpoint when `HF_INFERENCE_ENDPOINT_URL` is configured and falls back to local Transformers inference if that call fails; without an endpoint URL, it selects local inference directly. The deprecated generic `hf-inference` serverless model route is not used. Local loading is lazy and thread-safe, uses the repository's `tokenizer.json` through `PreTrainedTokenizerFast`, truncates inference

to 256 tokens, and selects CUDA when available unless the device is overridden. If the selected ViSoBERT path cannot produce a prediction, the versioned deterministic lexicon scorer is used. Every newly scored article stores the model/scorer name, the effective backend (local CPU/CUDA, dedicated endpoint host, lexicon, or empty-text handling), and a Boolean fallback flag.

Two point-in-time context variables are constructed. S_{decay} is the exponentially time-weighted mean of all valid dated-and-scored articles in the 90-day window, with a 30-day half-life and strict exclusion of post-cutoff observations. Rel_{sent} is the difference between the target's time-decayed score and the mean time-decayed score of related companies having at least three valid observations. Either value is marked unavailable when its observation requirement is not met; missing data is not encoded as neutral zero. The full compact observation set is used for these calculations, while the 15-article slice is display-only.

Neither variable is injected into α_2, any other registry formula, or factor ranking. They and the per-company related sentiment values are rendered in a separate prompt section that the LLM uses only for qualitative comparison with the OHLCV factor report.

In backtest mode, the live crawler is replaced by a pre-loaded historical sentiment store keyed by symbol and decision timestamp. Inclusion is based on timezone-aware `available_at`, not merely a publication date: date-only CafeF records become available at 00:00 on the next day in `Asia/Ho_Chi_Minh`, and undated records are rejected rather than used as fallback.

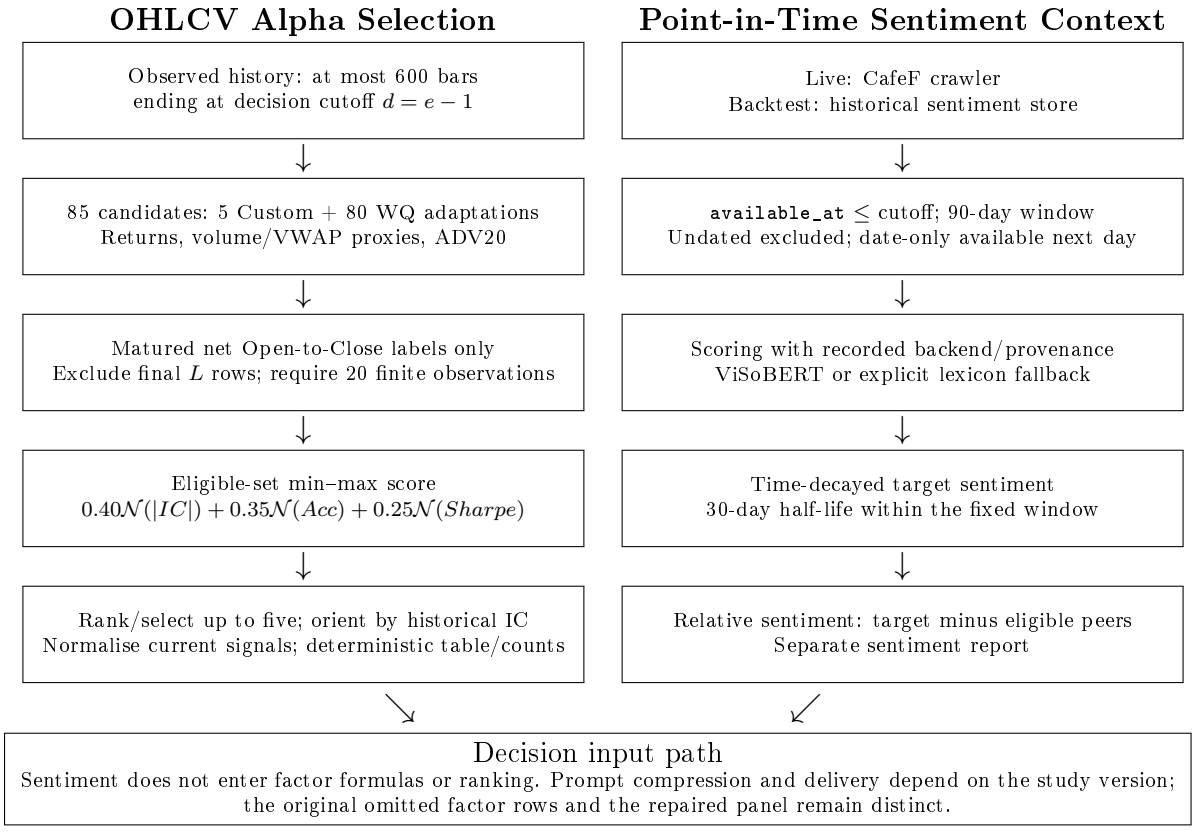


Figure 1: Internal data flow of the Alpha Agent

4.4. Pattern Agent

The Pattern Agent analyses a candlestick chart rendered from the current analysis window $[e - W, e)$ (45 candles in the main configuration). The chart is produced by `mplfinance` using a

custom colour scheme (green bodies for bearish candles and red bodies for bullish candles, following Vietnamese market convention) and encoded as a base64 PNG before graph execution.

The vision-language model is prompted to perform four sequential tasks. First, it identifies the dominant chart formation from a curated list of 16 classical patterns, such as head-and-shoulders inverse, double bottom, or various triangles and wedges. Second, it describes the three to five most recent candles in terms of body size, wick length, and relative position. Third, it assesses whether a breakout or breakdown is occurring. Finally, it commits to a directional bias (`bullish`, `bearish`, or `neutral`) alongside a confidence rating of `High`, `Medium`, or `Low`.

The prompt is explicitly structured to discourage non-committal answers and encourage a directional interpretation whenever sufficient evidence is present.

After inference, a deterministic verifier compares the reported bias with the final candlestick body and the direction of the last five closes. An obvious contradiction is appended to the report and downgrades its confidence. If the vision model fails or no chart image is available, the agent falls back to a text-only analysis of raw OHLCV values using the text LLM.

4.5. Trend Agent

The Trend Agent receives a separate chart in which support and resistance lines have been fitted algorithmically before LLM invocation.

Two trendline pairs are computed.

Close-price trend channel.. The first pair is derived from the closing price series alone. A linear regression line is fitted to all 45 candles in $[e - W, e)$, and the upper and lower deviations from this line identify the resistance and support pivots respectively. The slopes are then refined by minimising squared deviations subject to two constraints: the support line must not intersect any closing price from above, and the resistance line must not intersect any closing price from below.

High-low channel.. The second pair uses the high and low price series to identify the highest and lowest deviation points, producing channel lines that bound the full intra-bar range.

The support lines are drawn in blue, resistance lines in red, and rendered at 600 DPI to preserve legibility when compressed to base64.

The vision-language model is prompted to assess the slope of each support and resistance line, evaluate the distance between the lines (indicating channel width and price compression), determine the position of the most recent candles relative to each boundary, and produce a directional forecast for the applicable prediction horizon. The required JSON output fields encompass the trend direction, numerical support and resistance levels, slope characterisation, and overall confidence.

After inference, a deterministic verifier checks the reported trend against price relative to SMA20, the availability of SMA50, and the recent price slope. Contradictory claims are flagged and their confidence is reduced. If the vision-language model fails or no rendered chart is available, the agent falls back to a text-only analysis of raw OHLCV data, mirroring the strategy used by the Pattern Agent.

Scope of visual verification.. The two deterministic checks are *output-consistency gates*: they test whether the final directional statement contradicts a small set of OHLCV-derived facts. They cannot reveal which image region caused the model's answer, verify every claimed candle or trendline coordinate, or demonstrate sensitivity to the purported local pattern. A model that simply extrapolates the global trend can pass both gates whenever the shortcut and the rules agree. This distinction matters because observational financial charts couple local evidence with trend, making ordinary accuracy unable to identify grounding; controlled evidence interventions are needed to separate a

grounded responder from a trend-shortcut responder [11]. Accordingly, the evidence establishes reduced obvious contradiction, while audited visual faithfulness remains an open question. A direct audit would require matched charts that preserve global trend while altering only local candlestick semantics, complementary trend swaps that preserve the local pattern, and measurement of whether the model's direction changes in the rule-implied way. FinChart-Bench further cautions that current models remain unreliable on spatial and instruction-following aspects of financial charts [10].

4.6. Decision Agent

The Decision Agent synthesises all upstream reports into a single trading judgment. To remain within the Groq token-per-minute limit, each report is condensed before being passed to the prompt: the Indicator report is trimmed to its consensus summary block, the Alpha report retains only the factor table and LLM reasoning paragraph, and the Sentiment report drops the per-article listing.

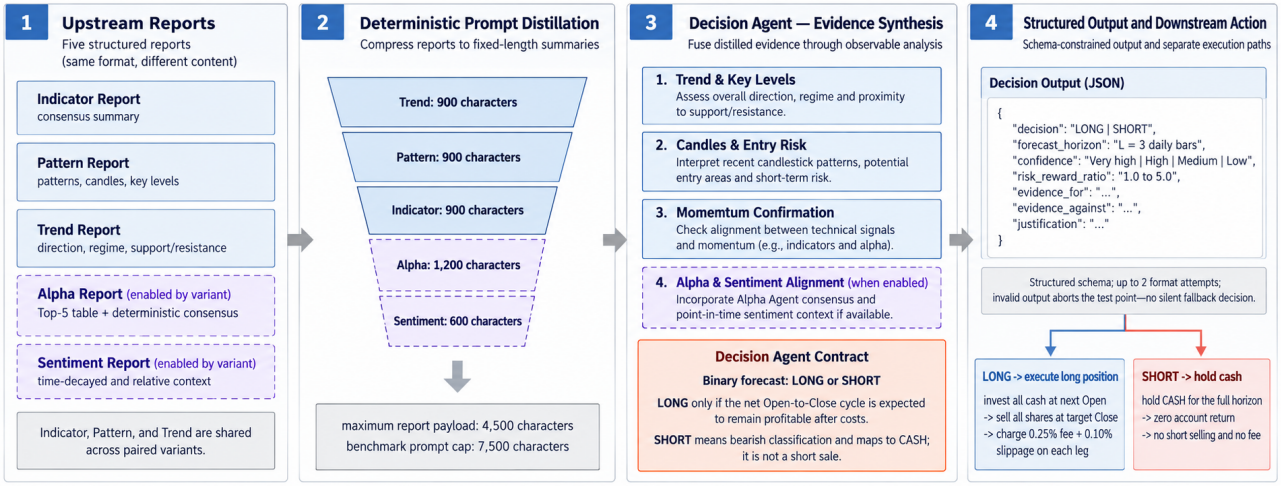


Figure 2: Decision Agent reasoning pipeline

The agent is prompted to reason through four sequential questions in a structured chain-of-thought process. It evaluates whether the price is near support or resistance while identifying the primary trend, checks if recent candles show reversal or continuation signals to ensure a low-risk entry, verifies whether momentum/trend indicators such as RSI and MACD confirm the price direction, and finally assesses if the quantitative alpha signals and sentiment align with or contradict the technical picture.

Following this reasoning, the model must output a JSON object comprising a forced binary direction forecast (`LONG` or `SHORT`), the `forecast_horizon`, the `confidence` level, a `risk_reward_ratio` clamped to $[1.0, 5.0]$, supporting and opposing evidence, and a final `justification` paragraph. A separate prompt instruction distinguishes this forecast label from an executed order.

The forced binary constraint supports direction classification only. Downstream account logic independently maps predictions to actions: `SHORT` maps to `CASH`, whereas `LONG` invests all available cash at the next open and closes the position at the configured target close after applying costs on both legs.

5. Experimental Setup

5.1. Dataset and Point-in-Time Data Contract

The daily evaluation universe contains nine Vietnamese equities spanning five economically distinct groups: information technology (FPT and CMG), banking (VCB and MBB), retail (MWG), food and beverages (VNM and BHN), and aviation and tourism (HVN and VJC). The purposive stratification combines the VN30/large-cap names specified in the frozen design (VCB, FPT, VNM, MBB, and MWG) with mid-cap or ownership-specialised firms (CMG, BHN, and HVN) and a large private airline (VJC). It was chosen to expose the agents to heterogeneous liquidity, ownership, and news environments; it is not claimed to be a random or market-representative sample. The benchmark contains 20 chronologically ordered test points per symbol, for 180 paired observations in total. Section 7 discusses the resulting external-validity limit, and Appendix Table F.40 reports the complete ticker and sentiment-coverage audit.

OHLCV observations are obtained through the project's `vnstock`-based data loader. At an exclusive test boundary e, every agent sees only bars in $[e - W, e)$, so the last visible decision bar is $d = e - 1$. Dynamic alpha selection receives a cutoff-safe prefix ending at the same decision bar and containing at most 600 observations. Neither the factor calculations nor the chart renderers can inspect the entry bar or target bar. The numerical, candlestick, and trend inputs use the same requested window length and fail closed when a complete window is unavailable.

The historical sentiment store applies the same cutoff. Only records with a parseable publication date no later than d are eligible; undated records are rejected in research mode instead of being borrowed from a later cache state. Across the supplied caches, all 180 articles have a parseable date. The mean number available in a 90-day point-in-time window is 5.77, and 73 of 180 test points (40.56%) contain at least eight articles. The legacy cache schema does not retain the scorer identity or fallback flag, so 0 of 180 articles can be verified as ViSoBERT-scored, 0 can be verified as lexicon-scored, and all 180 are conservatively reported as `legacy-unknown`. These figures measure verifiable provenance rather than infer model identity from confidence or sentiment values. The sentiment pathway remains text context for the Decision Agent and does not enter any OHLCV alpha formula.

5.2. Walk-Forward Protocol

All reported experiments use daily observations. The main configuration is $W = 45$, step size $S = 3$, and lookahead $L = 3$. For boundary e, the simulated entry is the next open O_e and the target exit is C_{e-1+L}. The realised net return of a candidate LONG cycle is

$$r_{e,L}^{\text{net}} = \frac{C_{e-1+L}}{O_e} \frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1, \tag{40}$$

where $c_{\text{fee}} = 0.0025$ and $c_{\text{slip}} = 0.001$ are applied separately on both the BUY and SELL legs. The economic direction label is `UP` only when $r_{e,L}^{\text{net}} > 0$ and is `DOWN` otherwise. This definition makes the classification target, alpha-selection target, and account return share the same Open-to-Close horizon and cost convention. Figure 3 summarizes the timing contract, per-cutoff processing sequence, paired shared-report comparison, and separate account ledgers.

5.3. Paired Baseline and Disentangled Ablations

The legacy benchmark compares Full with a control historically named No-Alpha, here called Base reports. The control removes both OHLCV factors and sentiment. The implemented paired shared-report protocol intends to deep-copy Indicator, Pattern and Trend reports within each pair; complete historical reports and requests have not been recovered to independently certify every

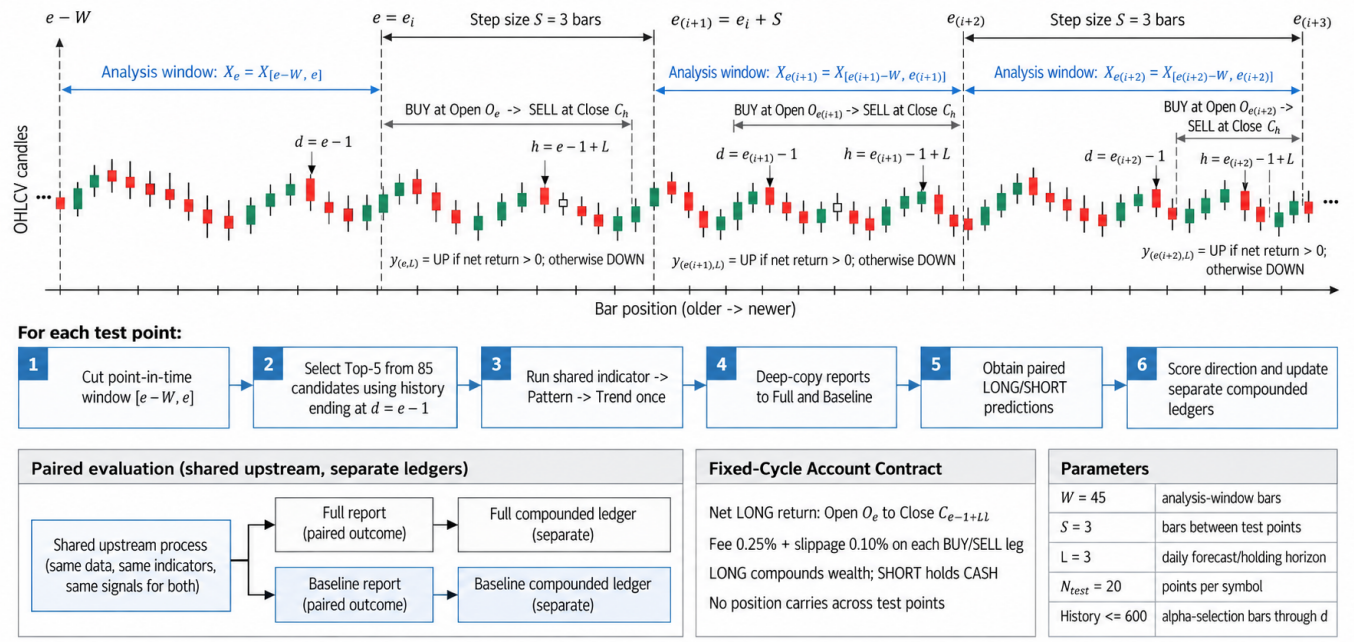


Figure 3: Walk-forward evaluation protocol.

Table 3: Main walk-forward configuration.

Parameter	**Value**	**Operational meaning**
Analysis window W	45 bars	Exact agent input $[e - W, e)$
Step size S	3 bars	Separation between consecutive test boundaries
Lookahead L	3 bars	Entry at O_e and exit at C_{e-1+L}
Test points	20 per symbol	180 paired observations over nine symbols
Alpha history	Up to 600 bars	Cutoff-safe calibration/evaluation prefix ending at $d = e-1$
Alpha registry	85 candidates	Top five selected at each test point
Random seeds	42	Python/NumPy and bootstrap reproducibility

archived pair. All 180 saved pairs contain valid binary decisions. Their accuracy difference is a Full–Base bundle gap, not an estimate isolating alpha information.

The legacy four-way experiment on FPT, VNM and VCB is intended to compare Full, Alpha-only, Sentiment-only and Base reports with shared upstream inputs. Its branch provenance remains unresolved, so its stored outcomes do not establish independent component contributions. The historical summary below measures improvement over the best single component, not a factorial interaction:

$$\mathrm{Gap}_{\text{best-single}} = \mathrm{Acc}_{\text{Full}} - \max(\mathrm{Acc}_{\text{AlphaOnly}}, \mathrm{Acc}_{\text{SentimentOnly}})\,. \tag{41}$$

5.4. Account and Statistical Metrics

Each variant starts with VND 50,000,000 and is simulated independently. Every LONG uses all available cash at O_e, pays fee and slippage on purchase, sells all shares at C_{e-1+L}, and pays both costs again on sale. Every cycle therefore ends in cash before the next test point. A SHORT forecast

Table 4: Paired benchmark variants.

Component	Full System	Base reports
Indicator report	Shared upstream snapshot	Identical shared snapshot
Pattern and Trend reports	Shared upstream snapshot	Identical shared snapshot
OHLCV alpha factors	Enabled	Disabled
Vietnamese sentiment context	Enabled	Disabled
Final Decision Agent	Receives all enabled reports	Receives only shared upstream reports

is a directional prediction only and maps to CASH because uncovered short selling of the underlying equity is disabled. With cycle return $q_k = r^{\text{net}}_{e_k,L}$ for LONG and $q_k = 0$ for SHORT, wealth compounds as

$$W_{k+1} = W_k(1 + q_k), \qquad R_{\text{acct}} = \frac{W_K}{W_0} - 1. \tag{42}$$

The legacy files store Sharpe calculated with $\sqrt{252}$ on cycle returns; this frequency convention is under methodological correction and is not used here to support an investment claim. Their drawdown is cycle-end rather than daily mark-to-market. The new numerical ledger additionally marks holdings at every observed session close, records daily quoted-notional turnover divided by prior equity, and separates held-session fraction from close-of-day exposure. Continuous holding buys once at the first entry and sells once at the final exit, including inter-cycle gaps. It is a price-series comparator, not a verified total-return account: corporate-action/cash-dividend ledgers are unavailable. Fractional shares, immediately reusable cash, zero interest, and the stated two-sided costs remain idealised research assumptions, not a venue-calibrated execution model.

Legacy McNemar, bootstrap, Newey–West and cross-symbol Wilcoxon outputs are retained as historical diagnostics with their original assumptions. They are not used to assert general significance, since 20 observations per symbol and common market dates do not constitute nine independent replications. The numerical reconstruction is descriptive and reports no superiority test. Balanced accuracy is the primary predictive summary, with MCC, macro-F1, raw accuracy, class support and both classes' precision/recall/F1 also reported. Undefined denominators are NA rather than zero. Pooled metrics aggregate confusion counts; macro-over-symbol metrics weight each symbol equally and require each symbol's metric to be defined.

5.5. Exploratory Numerical Baselines and Comparability

The additional numerical evaluation freezes a newly retrieved VCI price series through 30 September 2026 and retains all 20 legacy cutoff dates for each symbol. This is a reconstruction, not an authenticated copy of the historical LLM input snapshot. Canonical entry/exit prices and net labels are compared before any Full comparison is permitted. The 74 mismatched endpoints and two changed labels prevent that comparison, even though pooled class counts happen to remain 56 UP and 124 DOWN.

Training decisions end in 2025; validation begins in January 2026 and ends before each symbol's first test decision. Every training label exits strictly before the first validation decision, and every validation label exits strictly before the first test decision. Samples use three-session spacing and 300 warm-up sessions. An invalid BHN OHLC row on 13 September 2022 has no verified replacement, so the numerical feature/training view starts at the valid suffix after that row; the raw file and all test dates remain unchanged.

Table 5: Evaluation metrics and estimands.

Metric	Definition
Balanced accuracy	Mean of UP and DOWN recalls; both class supports must be nonzero.
MCC and class metrics	Matthews correlation; macro-F1; class supports and precision/recall/F1, with undefined values reported as NA.
Directional accuracy	Fraction of valid LONG/SHORT forecasts matching the net economic direction in Eq. (4).
Legacy bundle gap	Full accuracy minus Base reports accuracy; not isolated alpha information value.
Account return	Compounded fixed-cycle return $W_K/W_0 - 1$; LONG executes and SHORT holds cash.
Account Sharpe	Mean cycle return divided by population standard deviation and annualised by $\sqrt{252}$.
Maximum drawdown	Largest peak-to-trough decline on the compounded cycle-end equity curve.
Paired inference	Exact McNemar test, moving-block 95% lift interval, Newey–West mean test, and cross-symbol Wilcoxon signed-rank test.

Baselines are constant DOWN/CASH, always-LONG cycles, train-fitted majority, fixed 20-session momentum, logistic regression on price/indicator features, logistic regression on the same price features plus 85 factors and verified sentiment/missingness, and equal factor voting with train-only orientation. Five factor formulas containing whole-prefix normalisation receive explicit expanding past-only adapters in the new baseline pipeline; the legacy registry is unchanged. The broader factor set, absent peer context and missing reports mean that the second logistic is an information-family comparator, not a verified reconstruction of Full's exact numerical/textual inputs. Unknown historical scorer provenance makes sentiment unavailable rather than neutral.

Median imputation, missingness indicators and standardisation are fitted only on train during selection. The fixed regularisation grid is $C \in \{0.01, 0.1, 1, 10\}$ and threshold grid is $\{0.5, 0.4, 0.6\}$, selected by validation balanced accuracy with fixed-order ties. Selected preprocessing and logistic models are then refitted on train plus matured validation. A seeded random control uses the training momentum LONG rate; it is explicitly not a Full-exposure-matched control, since historical Full train predictions are missing. Causal per-symbol regimes use the close/SMA60 gap (above 2%, below −2%, or otherwise sideways). One deterministic numerical run is reported; it is not substituted for repeated LLM runs or multi-year LLM folds.

5.6. New Frozen Common-Snapshot Study

The primary revision evidence is a new historical exploratory study, not a relabelled reconstruction of the legacy run. Its frozen bank contains 102 symbol–date units from nine equities and twelve decision dates. There are two dates in each block: 3 July/22 September 2023, 1 July/20 September 2024, 2 January/31 March 2025, 1 October/19 December 2025, 1 April/24 June 2026, and 1 July/23 September 2026. The first three blocks contain eight symbols; BHN is eligible only in the last three after the documented OHLC exclusion. Thus eight symbols contribute twelve units each and BHN contributes six. No date or checkpoint is selected using LLM outcomes. The machine-readable bank schedule, rather than the older 20-point template, defines every train, validation and test index.

Each unit uses the same frozen OHLCV, selected-factor features and sentiment availability for all numerical and LLM comparisons. The analysis window is 45 sessions. The net target retains entry at the next open, exit at the third subsequent observed-session close, 0.25% fee and 0.10% slippage on each leg. Preprocessing, train-majority decisions, factor-vote orientation and logistic regularisation/threshold selection use purged chronological train/validation data; label exits precede the next split. Price logistic uses 225 price and indicator features. Market-input logistic adds 432 selected-factor, sentiment and missingness features from the same numerical input bank. It is an information-family comparator: numerical inputs are common, but neither chart pixels, report wording nor pretrained model knowledge is identical to the LLM representation. Fixed momentum, train-majority, constant CASH, always-LONG cycles and factor voting complete the classification controls. Continuous hold buys once and sells once within each symbol–block account and is an economic comparator, not a cycle classifier. No control is fitted to Full's test LONG rate; no Full-exposure-matched superiority claim is made.

For this new bank, 154 eligible text excerpts from an audit of 180 records (177 URL/text identities) were rescored locally using the pinned `5CD-AI/Vietnamese-Sentiment-visobert` revision recorded in the bank. Scores use $P(\text{POS}) - P(\text{NEG})$, a 256-token cap, date-only availability from the following day, deduplication, a 90-day window and a minimum of three articles; unavailable sentiment remains missing. Peer context is empty for all variants. This authenticates the new scoring process, not historical publication/content availability or financial-domain validity. Legacy sentiment remains unknown.

The main comparison is B (Base reports) versus AS (Full), with the same Indicator, Pattern and Trend reports per unit. B removes factors and sentiment. A preselected 18-unit panel evaluates B/A/S/AS/P three times with fixed shared upstream; A and S enable only factors and only sentiment, respectively. P retains sentiment but replaces the selected factor vector with a preselected train-only past donor. A separate nine-unit panel regenerates upstream three times for B/AS. The base realization is shared with the main study, not called twice. These contracts require 474 research decisions; ten smoke decisions are outside evaluation. No incomplete job, fallback output or synthetic CASH prediction enters the release.

New text/vision calls use `openai/gpt-oss-20b` and `qwen/qwen3.8-27b`, configured temperature zero and output caps 2,048/1,024, respectively. Requests, raw responses, parent hashes and runtime/source identities are archived. An author-authorised operational amendment replaces internal budget and deadline stops with routing across five provider keys; model, prompt, units, variants and targets remain frozen. There are 175 research decisions before and 299 after this amendment, with one main pair crossing epochs. This is disclosed rather than assumed to be an immutable provider revision. There are 756 attempts, 734 successful responses and 22 archived errors including smoke; 1,229,635 prompt and 420,761 completion tokens are SDK-observed successful usage, excluding unknown failed-call usage. No monetary cost is inferred.

Class support, both-class metrics, pooled and equal-symbol macro summaries, and strata by symbol, block and causal regime use common support. Regimes use the decision-close gap from its trailing SMA60: above 2% is bull, below -2% bear, and otherwise sideways. For paired BA contrasts, all symbols in each block are resampled together over six blocks, with 10,000 bootstrap replicates and seed 20261102. The resulting few-block intervals are descriptive. Three-repeat dispersion is restricted to the selected panel, not assigned to the 102-unit one-pass study. The economic ledger uses fractional shares and immediate cash reuse. Means over 51 separate symbol–block accounts are not a joint portfolio, a certified settlement simulation or a dividend-adjusted total-return estimate.

5.7. Temporal Coverage and Dependent Panel Audit

The temporal audit reuses the complete frozen prediction bank; it makes no new model calls or outcome-dependent changes to dates, units or predictions. Tables A.24 and A.25 detail its support. The period extends from 3 July 2023 to 23 September 2026, but is sparsely sampled: twelve decision dates and 36 distinct future target-session dates, not continuous daily evaluation across those years. There are 306 eligible symbol–target-session observations, which share calendar shocks and cannot be counted as independent replications. Separately funded account simulations likewise do not imply statistical independence.

The snapshot contains 1,681 observed vendor sessions per symbol from 2 January 2020 through 30 September 2026. A contradictory BHN bar on 13 September 2022 requires the already frozen valid-suffix rule, excluding 673 prefix rows while preserving their raw bytes. Its remaining history does not meet the 600-session warm-up in the first three blocks. The panel therefore has 102 eligible cells out of 108 date–symbol slots; the six missing BHN cells remain missing for every classifier. Predictions are joined on actual calendar date, symbol and variant, not a within-symbol test index. The full nine-classifier grid has 972 slots with 54 explicit missing cells.

All 51 outer symbol–block scopes have expanding past training samples and chronological inner validation. Training label exits precede the first validation decision; validation label exits precede the first test decision. The first train cutoff has at least 600 observed history sessions. Depending on symbol and block, there are 32–318 training samples and 18–21 validation samples. Stored logistic imputation and scaling statistics were independently recomputed from exactly the purged training plus matured validation inputs and matched in every scope; classifiers were not refitted. Hyperparameter selection follows the frozen validation procedure. The LLM's adaptive factor selector remains the frozen causal 600-session policy at each cutoff, not a newly nested factor-discovery experiment; multiplicity and selector stability remain separate requirements.

The original trend context is the close/SMA60 gap. To describe coverage more fully, two post-inference descriptive dimensions are added: trailing 20-session close-return sample volatility and trailing 20-session mean Close×Volume. Each high/low threshold is its symbol–fold median on train cutoffs only. These values use no future bars or test-label fitting, but the new stratification itself was not preregistered before inference. The turnover proxy retains vendor quote/volume units and is not a measure of spread, depth or executable liquidity. Context labels describe individual price paths, not an independently verified market-index regime.

Group metrics report unit, class, symbol and distinct-date support alongside LONG frequency. Exposure is checked against the frozen daily ledgers over the three target sessions per unit; its denominator excludes the long cash gaps between selected decisions. It is not a calendar-wide portfolio exposure. Whole-block bootstrap draws keep all eligible symbols, both dates and every variant together, preserving contemporaneous dependence and the missing mask. Coverage-cohort and leave-one-block-out sensitivities are descriptive; three or six blocks do not establish inter-block independence or reliable small-sample precision. The cross-symbol legacy Wilcoxon value is not used as an inferential test for this study.

This remains a purposively selected surviving nine-symbol universe. There is no authenticated point-in-time constituent, listing/delisting or suspension registry; zero volume does not certify suspension. All labels matured by 28 September 2026, before protocol freeze on 2 November 2026, so there are zero prospective test units. The served text/vision weight revision and training-corpus exclusion are unverified. Archived model names, release dates, temperature and request fingerprints do not prove absence of memorised historical prices. No contamination-free, market-wide or untouched prospective performance claim is made.

5.8. Matched Repeated Executions

The repeated-run audit conditions on the two panels chosen before inference: 18 units (six UP/twelve DOWN, eight symbols, five dates, three blocks) have three Decision executions for each B/A/S/AS/P variant with upstream fixed; nine units (two UP/seven DOWN, nine symbols, three dates, three blocks) have three regenerated upstream realizations for B/AS, each followed by one Decision execution. The base realization is reused from the main comparison by design. Only one unit overlaps the panels, sharing its two r0 B/AS decisions; their union contains 322 unique decision executions. Each distinct execution has its own successful raw request ID. The 90 fixed-upstream unit–variant groups retain identical actual prompt hashes across all three Decision calls. Hash identity establishes sameness, not completeness of the planned content; Section 5.9 audits that content. Every variant in a realization has the same upstream parent. Regenerated vision nodes have 88 distinct successful requests across the panel union. Repeated identical predictions are permitted, but copying a raw response into another repeat identity is rejected.

The configuration audit records cutoff/asset/prompt hashes, frozen source identity, model aliases, SDK metadata, provider key slots, timestamps and every failure. All 756 attempted calls record the same SDK temperature 10^{-8}: the pinned `langchain-groq` 1.1.2 adapter converts a configured zero to that value. This is observed client metadata, not an independent provider wire capture or weight-revision attestation. It corrects the shorthand "temperature zero" without implying deterministic hosted outputs. Output caps remain 2,048/1,024; no hosted decoding seed is supplied by the frozen caller. Source-byte hashes are authoritative where the model run's exact Git commit or served weight revision is unrecorded. The audit does not infer historical configuration from current defaults.

Within-job successful call order matches all 98 registered panel-related jobs. However, r0 B/AS originates in the earlier main pair and the remaining A/S/P r0 calls occur later; r1/r2 rotate variant order. Quota interruptions, provider routing and timing are observed confounds, not a randomized provider experiment or a complete Latin-square counterbalance. All 22 archived errors remain in the 756-attempt inventory; valid decisions are structured model outputs without synthetic CASH or model fallback. The 734 successful calls include ten smoke decisions and their upstream calls, excluded from metrics.

For each variant, accuracy, BA, MCC, macro-F1, LONG rate and account return are calculated separately for executions 0, 1 and 2. We report the mean, sample SD (ddof = 1), median, range and paired contrasts. Return is the mean of 13 decision-panel or nine upstream-panel symbol–block accounts containing only their selected cycles, under the frozen R1 costs and daily ledger. It is neither a pooled portfolio nor the broad 51-account return. Per-cutoff disagreement and the full three-prediction vector are retained.

The descriptive interval procedure is an analysis added after inference, not a preregistered precision guarantee. Each of 10,000 draws (seed 20261102) resamples whole temporal blocks with their complete cross-section and also resamples the three execution indices, using the same draws for all variants. The statistic is the mean of the sampled per-execution scores, not pooled classification of $3N$ independent markets. Undefined class/dispersion draws are counted and intervals use only finite draws; all metric-specific counts are published. With only three runs and three blocks per panel, coverage is not calibrated. No best run, seed or checkpoint is selected and no ensemble is substituted for the main r0 result. The different panels and lack of Decision repeats within every upstream realization preclude a crossed variance-component decomposition.

5.9. Factorial Report and Placebo Content Audit

The same 18-unit, five-date, three-block Decision panel implements B=(0,0), A=(1,0), S=(0,1) and AS=(1,1), plus P with a training-donor alpha vector and the same sentiment as AS. All 18 units

meet the three-locally-scored-article gate; this is source coverage, not independent publication-PIT or financial sentiment validation. Missing-sentiment units are not used as sentiment effects. The original compressed-report and repaired full-value studies each have three fresh Decision executions per branch; they are analysed separately.

Actual requests, rather than only branch flags, are audited. All 54 cutoff–repeat groups share the same Indicator/Pattern/Trend parent hashes and the same three upstream text sections. A and AS share their alpha section; S, AS and P share their sentiment section. A contains no sentiment section, S contains no factor section, and the common priority rules still mention alpha and sentiment even for B. Thus these contrasts include report presence and resulting instruction/context effects; they do not isolate news information from prompt presence.

The audit uncovers an important implementation limitation. The legacy alpha distiller splits on the string `--`, which also occurs inside the Markdown table separator. It retains the table header and signal-count consensus but drops all five factor-value rows. All 162 alpha-bearing panel prompts, and all 264 alpha-bearing prompts among the 474 research decisions, have zero such rows. The feature bank's numerical vectors were not the full numerical content delivered to the Decision Agent. Prior "Full" and "alpha-only" names denote the implemented report branches, not proof of full factor-value transmission. Six of the 54 AS/P prompt pairs are identical after this compression, despite separate response IDs.

The donor map is reproducible from seed 20261102 plus the unit-ID hash. Each donor is in the purged training set; its three-session label exit precedes validation. The whole five-position past vector is reused jointly, not independently shuffled scalar by scalar, with matching schema and availability mask. Values, signals and skill fields are mapped by rank while current factor names are retained: 84 of 90 rank slots have a different donor name. This is a rank-vector perturbation, not a name-matched per-factor covariance control. After compression the historical P branch principally perturbs signal consensus, so it is not represented as the requested complete factor-value placebo.

For the original compressed-report study, the standard interaction and joint seven-contrast envelope were added after inference; exact earlier family registration is not authenticated. For the repaired study, the same seven BA contrasts, seed 20261102 and 10,000 block/run draws are explicitly locked before its calls. The family is AS–B, A–B, AS–S, S–B, AS–A, AS–P and the interaction in Eq. 38. Pointwise finite-draw percentiles and the joint maximum-absolute-deviation envelope are descriptive; three executions and three blocks do not establish calibrated simultaneous coverage or family-wise error control. Secondary endpoints are accuracy, MCC, macro-F1, both-class recall, LONG frequency and mean account return. Prior exposure to the historical sample precludes prospective confirmation.

The repaired bank contains 90 distinct templates and 270 registered fresh Decision jobs on the same 18 units. It reuses archived upstream reports and requires no new vision calls. The bank's timestamp precedes the first new request (03:35 versus 03:51 UTC on 3 November 2026), and all 576 source/runtime pins verify. All 274 attempted calls retain the locked text-model alias, SDK temperature 10^{-8} and 2,048 output-token cap. Four provider `json_validate_failed` rejections are retained as errors and retried; none becomes a prediction. All four rejections concern S jobs, so the reported distributions condition on eventually schema-valid outputs. All 270 distinct successful responses pass the native schema and checkpoint replay, with no model or synthetic CASH fallback.

The actual-request audit independently parses all five values, signals and skill fields from each of the 162 alpha-bearing successful prompts and matches them against the frozen treatment or past donor by rank. It confirms 54 paired groups with shared upstream text/parents/rules, matching A/AS alpha, matching S/AS/P sentiment, and AS/P differing only inside alpha content while retaining all five current names and the same schema/headings. Every within-group successful-call order matches the registered job sequence. Prompt lengths remain below 6,500 characters; exact

token-length equality or complete order counterbalancing is not claimed. Provider routing/timing and unverified served weights remain observational confounds.

No request ID or response is copied from the earlier study. The repaired renderer exposes the complete rank-vector control while preserving the old distiller, source archives and predictions byte-for-byte. It changes values, signals and skill metadata jointly, not one scalar in isolation; donor rank dependence is not named-factor covariance. Report-level sentiment contrasts still include section-presence effects. Neither a positive contrast nor repeated inference establishes isolated component information without these qualifications. Old and repaired distributions are not pooled or treated as a causal before/after estimate of the repair.

5.10. Implementation Defaults and Legacy Robustness Design

The legacy implementation description names text model `openai/gpt-oss-20b` and vision model `qwen/qwen3.8-27b`, configured temperature zero, output caps 2,048/1,024 and up to three client retries. These are described defaults, not authenticated per-call provenance for the archived legacy artifacts. The new study's actual client metadata and provider-quota retry policy are audited separately in Section 5.8. Python/NumPy seed 42 and legacy bootstrap seeds 42–44 concern local utilities, not hosted decoding. A nominal zero setting and a local seed do not establish bitwise-deterministic hosted outputs.

The configured provider identifiers are recorded, but the Groq endpoint did not expose an immutable weight digest or provider-side revision in the archived run artifacts, and the aliases were not pinned to such a revision. Exact replay across a later provider update is therefore not guaranteed. This limitation is separate from checkpoint reproducibility: calls are executed sequentially with rate-limit pacing, and resuming a checkpoint preserves completed outputs instead of regenerating them. The legacy sentiment caches have an analogous, explicitly reported provenance limit: their article scores predate scorer metadata and are labelled `legacy-unknown` rather than attributed retrospectively to a ViSoBERT revision. The corrected runtime records `scorer`, `scorer_backend`, and `is_fallback` for newly scored articles, but this software change neither identifies the backend of historical cache entries nor changes any reported benchmark value.

Table 6: Implementation defaults described with legacy experiments; not authenticated legacy per-call provenance.

Component	**Configuration**
Text / vision models	`openai/gpt-oss-20b` / `qwen/qwen3.8-27b`; temperature 0.0
Decoding controls	Text/vision caps 2,048/1,024 tokens; three client retries; hidden reasoning requested where supported
Randomness scope	Python/NumPy seed 42; bootstrap seeds 42–44; no hosted-decoding seed
Model revision	Provider aliases recorded; immutable Groq weight revision/digest unavailable and not pinned
Initial capital	VND 50,000,000 independently for each variant
Costs	0.25% fee and 0.10% slippage on each BUY and SELL leg
Position rule	LONG invests all cash for one fixed cycle; SHORT remains in CASH
Primary protocol	`paired_shared_reports`; 20 test points per symbol
Four-way protocol	`four_way_shared_upstream`; Full, Alpha-only, Sentiment-only, Baseline

The robustness experiment evaluates FPT and VNM with 20 test points per configuration. Panel A changes the common analysis window to $W \in \{30, 45, 60\}$. Panel B compares `zscore_tanh`,

`minmax`, and `rank` alpha normalisation. Panel C compares default, equal, and IC-heavy ranking weights. For Panels B and C, the No-Alpha branch is reused from the same shared upstream/control checkpoint within each symbol and panel; its decisions must therefore be invariant to parameters that only affect the Alpha branch. Panel A deliberately changes the shared input window, so its No-Alpha output is not expected to remain constant across rows.

6. Backtest Results

6.1. New Common-Snapshot Baseline Comparison

All 474 research decisions are complete, including all 102 main B/AS pairs, the 18-unit decision panel and the nine-unit upstream panel. Ten smoke decisions are excluded. The main support is 25 UP and 77 DOWN net outcomes; all nine classifiers use exactly these 102 units, with no excluded prediction. Tables 7 and 8 report the pooled comparison and both classes. Appendix A reports equal-symbol macro results, every baseline for each symbol, and block/regime strata; complete per-class/confusion counts for all 189 groups are also available in the machine-readable release.

Table 7: New common-snapshot comparison: 102 paired units, 25 UP and 77 DOWN net labels.

Variant	Acc.	BA	MCC	Macro-F1	LONG	Return
Full (AS)	62.75	48.31	-0.035	48.24	22.55	-0.88
Base reports (B)	57.84	47.77	-0.041	47.65	31.37	-1.03
Constant CASH	75.49	50.00	NA	43.02	0.00	0.00
Always-LONG cycles	24.51	50.00	NA	19.69	100.00	-3.00
Train majority	75.49	50.00	NA	43.02	0.00	0.00
Momentum (20)	49.02	47.32	-0.046	44.86	48.04	-1.64
Logistic price	67.65	55.61	0.114	55.68	23.53	-0.68
Logistic market inputs	60.78	52.42	0.044	51.82	32.35	-1.10
Factor vote	51.96	51.97	0.034	48.34	49.02	-1.46
Continuous hold	NA	NA	NA	NA	NA	-2.13

Percentages except MCC; NA denotes undefined or inapplicable. Return is the mean over 51 independent symbol–block accounts, each with two cycles. Continuous hold buys once and sells once per account; its classification metrics are inapplicable. No joint portfolio or total-return claim is made.

Full has TP/FN/TN/FP of 5/20/59/18 and Base 7/18/52/25. Full's 62.75% raw accuracy exceeds Base's 57.84%, but is below CASH and train-majority at 75.49%. Its BA is 48.31%, compared with Base at 47.77%, CASH at 50.00%, price logistic at 55.61% and market-input logistic at 52.42%. Full's MCC is -0.035; neither a significant forecasting improvement nor superiority over these controls is established. Full issues 23 LONG calls versus Base's 32. Its lower LONG frequency coincides with higher raw accuracy in a DOWN-heavy sample; this does not identify useful factor information. Full UP precision/recall is 21.74%/20.00%, and DOWN precision/recall is 74.68%/76.62%. Macro BA is 45.96% for Full and 50.07% for Base over eight defined symbols, unlike their pooled weighting; the macro table reports all defined denominators rather than filling undefined values with zero.

Full's mean account return is -0.88%, versus -1.03% for Base, 0.00% for CASH, -0.68% for price logistic, -3.00% for always-LONG cycles and -2.13% for continuous hold. These are means over 51 separate accounts, not a portfolio return. The small relative loss reduction over Base establishes neither profitability nor isolated alpha value. Every account has a daily mark-to-market ledger with costs, turnover, exposure and daily-close drawdown; the same execution assumptions apply throughout.

Table 8: Both-class pooled metrics on the same 102 units.

Variant	Class	Support	Predicted	Precision	Recall	F1
Full (AS)	UP	25	23	21.74	20.00	20.83
Full (AS)	DOWN	77	79	74.68	76.62	75.64
Base reports (B)	UP	25	32	21.88	28.00	24.56
Base reports (B)	DOWN	77	70	74.29	67.53	70.75
Constant CASH	UP	25	0	NA	0.00	0.00
Constant CASH	DOWN	77	102	75.49	100.00	86.03
Always-LONG cycles	UP	25	102	24.51	100.00	39.37
Always-LONG cycles	DOWN	77	0	NA	0.00	0.00
Train majority	UP	25	0	NA	0.00	0.00
Train majority	DOWN	77	102	75.49	100.00	86.03
Momentum (20)	UP	25	49	22.45	44.00	29.73
Momentum (20)	DOWN	77	53	73.58	50.65	60.00
Logistic price	UP	25	24	33.33	32.00	32.65
Logistic price	DOWN	77	78	78.21	79.22	78.71
Logistic market inputs	UP	25	33	27.27	36.00	31.03
Logistic market inputs	DOWN	77	69	76.81	68.83	72.60
Factor vote	UP	25	50	26.00	52.00	34.67
Factor vote	DOWN	77	52	76.92	51.95	62.02

Precision, recall and F1 are percentages. Support counts outcomes; Predicted counts calls for that class. Precision with no predicted class is NA.

Table 9: Paired balanced-accuracy contrasts, percentage points.

Contrast	Difference	Descriptive 95% interval
Full minus Base reports (B)	0.55	[-5.91, 8.00]
Full minus Constant CASH	-1.69	[-8.81, 5.16]
Full minus Always-LONG cycles	-1.69	[-8.81, 5.16]
Full minus Factor vote	-3.66	[-16.67, 9.33]
Full minus Logistic market inputs	-4.10	[-22.22, 7.15]
Full minus Logistic price	-7.30	[-18.70, 1.79]
Full minus Momentum (20)	0.99	[-21.11, 17.23]
Full minus Train majority	-1.69	[-8.81, 5.16]

10,000 joint cross-sectional six-block bootstrap resamples, seed 20261102. Only six historical blocks are available; intervals are descriptive and are not a power or superiority guarantee.

Full–Base BA differs by only 0.55 percentage points; its descriptive six-block interval spans −5.91 to 8.00 percentage points. Full minus price-logistic BA is −7.30 points; its interval runs from −18.70 to 1.79 points. No superiority claim is based on these few-block intervals. A separate gross-return diagnostic changes support to 32 UP/70 DOWN but keeps the same predictions elicited for the net target; Full gross BA is 47.23%. It is not a newly trained gross-direction model or a replacement target.

Table 10: Preselected repeated panels: three executions per variant.

Panel	Variant	Units	Mean BA	SD (pp)	Mean LONG	Disagree
Decision	B	18	25.00	4.17	50.00	11.11
Decision	A	18	30.56	6.36	42.59	14.81
Decision	S	18	27.78	2.41	46.30	7.41
Decision	AS	18	30.56	10.49	37.04	22.22
Decision	P	18	33.33	4.17	38.89	7.41
Upstream	B	9	90.48	4.12	37.04	7.41
Upstream	AS	9	97.62	4.12	25.93	7.41

Decision panel: 18 units, eight symbols, five dates, three blocks; shared upstream is fixed. Upstream panel: nine units, nine symbols, three dates, three blocks; upstream is regenerated. Disagree is mean pairwise prediction disagreement in percent. Dispersion applies only to each selected panel, not to all 102 units.

The selected decision panel is substantially worse than the broad study: Full mean BA is 30.56% with sample SD 10.49 points and 22.22% pairwise prediction disagreement. Its BA advantage over B across repeats is $+4.17, -8.33, +20.83$ points; versus placebo P it is $-4.17, -8.33, +4.17$ points. These mixed contrasts do not establish factor or sentiment information value. The upstream panel has Full mean BA 97.62% versus Base 90.48%, both with SD 4.12 points, on only nine units/three dates. Neither selected panel is pooled with the main study or used to choose a preferred checkpoint.

6.2. Temporal and Coverage-Cohort Sensitivity

The frozen sample covers rising, falling and sideways individual price contexts rather than assigning a regime from the name of a year: there are 33 bull, 35 bear and 34 sideways units (Table A.26). Full BA is respectively 50.00%, 41.67% and 48.22%, with LONG frequencies 33.33%, 14.29% and 20.59%. These strata overlap in calendar dates and are not independent market replications. The early-2025 block contains 16 DOWN and zero UP net outcomes; its class-balanced score is undefined, not filled with zero or omitted from the overall panel. Volatility and turnover-proxy strata are reported as post-inference diagnostics; favorable subsets are not selected to define a new trading rule or claim.

Table A.27 separates unequal coverage from broad pooled weighting. On the common eight-symbol/six-block cohort ($N = 96$, twelve dates), Full/Base BA is 48.61%/46.53%; the paired difference is 2.08 points with descriptive interval $[-4.10, 8.67]$. On nine symbols in three recent blocks ($N = 54$, six dates), BA is 47.18%/46.67%, difference 0.51 points with interval $[-8.33, 6.25]$. Neither cohort replaces the planned 102-unit result; they overlap and are not pooled as extra observations.

Omitting one of the six blocks changes Full–Base BA from −2.20 to +2.75 percentage points. Full remains below price-logistic BA in all six leave-one-block-out calculations (differences −11.11 to −2.63 points), but these sensitivities are not independent significance tests. The sparse historical panel broadens temporal coverage without establishing a stable cross-regime advantage, statistical power or external validity. All targets precede protocol freeze, and the observed provider metadata cannot resolve training contamination for either model.

6.3. Matched Repeat Distributions

Tables A.28–A.31 report the three-execution distributions on each fixed support. On the 18-unit Decision panel, Full accuracy is 37.04% with SD 11.56 points; BA is 30.56% with SD 10.49 points, median 29.17% and range 20.83–41.67%. Mean MCC is −0.378 with SD 0.195. Full–Base BA differences are $+4.17, -8.33, +20.83$ points: mean 5.56, SD 14.63 and median 4.17 points. The conditional two-level interval is $[-9.56, 23.08]$ points and includes both signs. These variations do not support a reliable bundle advantage. Full mean return is −1.544% (SD 0.679 points) over its 13 panel accounts, versus Base at −1.786% (SD 0.239). The paired return difference ranges from −0.595 to +1.224 points. Lower mean loss does not establish profitability or isolated factor information value.

On the separate nine-unit upstream panel, Full BA is 97.62% with SD 4.12 points, versus Base 90.48% with SD 4.12. Its Full–Base BA difference is 7.14 points in each realization, but the conditional block/run interval is $[0.00, 14.29]$. Mean returns are +0.624% for Full and +0.284% for Base over nine accounts. This much stronger selected support is not pooled with the Decision panel or used to choose a favorable run. Only one market unit overlaps, so subtracting the two panels' SDs cannot identify an upstream variance contribution. Full pairwise disagreement is 22.22% within the Decision panel and 7.41% within the upstream panel; these are support-specific descriptions, not a controlled comparison of stages.

Of 10,000 block/run draws, 388 lose class support for Decision-panel BA and 3,014 for upstream-panel BA. The reported BA intervals therefore use 9,612 and 6,986 finite draws, respectively. Missing metrics are not filled with zero, and the resulting conditional percentiles are not claimed to have calibrated 95% coverage. Machine-readable outputs retain every run, cutoff disagreement, metric-specific undefined count and daily panel ledger. The observed dispersion conditions on the fixed inputs and recorded client configuration; unknown served weights and provider/timing changes prevent attributing it solely to intrinsic sampling noise.

The 24-row legacy default inventory covers twelve source artifacts for FPT, VCB and VNM. For each ticker, nominal defaults span two recorded cutoff and cycle-endpoint families. Even within one family, complete reports, prompts, code identity and independent executions are not authenticated. Their 15–20-point spread is not treated as an estimate of stochastic variance under identical inputs. The old window/normalisation/weight sweeps are not newly replicated here; their robustness and component-attribution claims remain withdrawn, and unresolved branch provenance is assessed separately.

6.4. Corrected Value-Control Panel and Attribution Limits

Tables A.33 and A.34 describe the completed repaired panel, not the original compressed-report runs. Across three executions on the same 18 units, Full has mean accuracy 53.70%, BA 44.44% (SD 2.41 points), MCC −0.121 and LONG frequency 24.07%. Base BA is 29.17%, A 40.28%, S 26.39% and P 37.50%. CASH accuracy is 66.67% with BA 50.00%, so the positive bundle contrast does not establish superiority over the constant reference.

Mean BA differences are AS–B +15.28, A–B +11.11, AS–S +18.06, S–B −2.78, AS–A +4.17 and AS–P +6.94 points. AS–P varies as $+4.17, +12.50, +4.17$ points (sample SD 4.81); its pointwise descriptive interval is $[2.08, 12.82]$. However, the predefined seven-contrast joint envelope is $[-6.55, 20.44]$. Its radius is 13.49 points, based on 9,612 jointly finite draws and 388 class-loss draws out of 10,000. The AS–B and AS–S joint envelopes are respectively $[1.79, 28.77]$ and $[4.56, 31.55]$; these too are descriptive, not calibrated significance findings. The standard interaction is +6.94 points (SD 6.36), with pointwise interval $[-8.97, 16.67]$; it does not establish positive synergy.

All 15 variant–execution combinations correctly identify only one of six UP cycles (recall 16.67%). Full/P DOWN recalls average 72.22%/58.33%, and their LONG frequencies are 24.07%/33.33%.

Thus the measured AS–P BA gain comes entirely from fewer DOWN errors, accompanied by lower LONG exposure, rather than additional correctly identified profitable cycles. Full mean net account return is −0.670% (SD 0.055 points), versus −1.121% for P and −1.640% for Base; the paired AS–P return gap is +0.451 points (SD 0.315; interval $[0.000, 0.796]$). Returns average 13 panel symbol–block accounts per execution, not a portfolio. All variants have negative mean returns, below CASH at zero. These results do not support profitability, exposure-independent alpha or isolated sentiment information. Per-class support and equal-symbol macros with their defined denominators are published separately from the pooled-per-execution scores.

Table A.35 verifies complete factor delivery and genuine new responses. The older compressed-report contrasts remain in Table A.32; their AS–P and interaction means were both −2.78 points and are not substituted for, pooled with or selectively replaced by the repaired results. Different inference times and content preclude interpreting their change as a causal repair effect. The completed control tests a whole training-donor rank-vector perturbation under the same report structure, with its mapping and precision limits explicitly retained; it does not turn the observed conditional advantage into a general factor-information claim.

6.5. Legacy Nine-Symbol Paired Benchmark

The original 180 net labels contain 56 UP and 124 DOWN outcomes. Full has confusion counts TP/FN/TN/FP of 16/40/101/23, compared with 15/41/77/47 for Base reports. Its 65.0% accuracy is below constant DOWN/CASH at 68.9%. Full's balanced accuracy is 55.01% and MCC is 0.113; the Base reports values are 44.44% and −0.108. Full issues 39 LONG calls versus 62 for Base reports. Of 25 additional correct Full decisions, 24 are DOWN and one is UP. This pattern is consistent with fewer LONG decisions in a predominantly unprofitable-cycle sample, but does not identify the cause as alpha information. Tables 11 and 12 report the imbalance-aware audit without suppressing the stronger constant baseline.

Table 11: Legacy class-imbalance audit on the original 180 labels and cycle endpoints.

Variant	N	Acc. (%)	BA (%)	MCC	Macro-F1 (%)	LONG (%)	Mean return (%)
Full (legacy)	180	65.00	55.01	0.113	54.96	21.67	-1.68
Base reports (legacy)	180	51.11	44.44	-0.108	44.53	34.44	-8.43
Constant DOWN / CASH	180	68.89	50.00	NA	40.79	0.00	0.00
Always-LONG cycles	180	31.11	50.00	NA	23.73	100.00	-17.69

Predictive metrics are pooled; return is the mean of nine independent accounts, not a pooled portfolio. UP/DOWN support is 56/124. Undefined values are NA. Always-LONG closes and reopens each cycle; it is not buy-and-hold.

Table 13 retains the historical paired summaries for traceability. Its Full–Base gap is a bundle comparison. Stored cross-symbol Wilcoxon and small-sample intervals are descriptive legacy diagnostics, not evidence of a general significant alpha or forecasting improvement.

Full's recomputed mean account return is −1.6833%; averaging the already rounded legacy percentages instead gives −1.6856%, explaining the historical −1.69% display. Both are below CASH at zero. These are means of nine independent accounts, not a joint portfolio. Correct SHORT predictions earn no return under the long-only map. Neither the Full–Base accuracy gap nor relative loss reduction establishes isolated alpha value or profitability.

6.6. Numerical Reconstruction and Daily Ledgers

Tables 14 and 15 report the additional numerical study under Section 5.5. These results belong to the new snapshot and are not ranked against Full's legacy outcomes. The price-only and factor-augmented logistic models answer different information-set questions; neither has verified identical

Table 12: Per-class legacy precision, recall and F1, in percent.

Variant	Class	Support	Precision	Recall	F1
Full (legacy)	UP	56	41.03	28.57	33.68
Full (legacy)	DOWN	124	71.63	81.45	76.23
Base reports (legacy)	UP	56	24.19	26.79	25.42
Base reports (legacy)	DOWN	124	65.25	62.10	63.64
Constant DOWN / CASH	UP	56	NA	0.00	0.00
Constant DOWN / CASH	DOWN	124	68.89	100.00	81.58
Always-LONG cycles	UP	56	31.11	100.00	47.46
Always-LONG cycles	DOWN	124	NA	0.00	0.00

A class with support but no predicted positives has undefined precision and zero recall/F1. All symbol-level confusion matrices and class metrics are retained in the reanalysis CSV/JSON.

archived Full reports. The continuous-hold account buys once and sells once, unlike always-LONG cycles. Daily-close drawdown, turnover, fees and exposure are retained per symbol in the reanalysis ledgers; no daily Full ledger is inferred from mismatched reconstructed prices. The train-rate random control is not represented as matching Full's exposure. Accordingly, the newly implemented baselines alone do not establish a valid comparison with legacy Full. The newly frozen study in Section 6.1 supplies the actual common-snapshot LLM comparison, as historical exploratory evidence.

6.7. Legacy Robustness and Sensitivity

Table 16 reports all six completed sweeps. Each row contains 20 paired test points. Panel A changes the input window for both variants and therefore changes the No-Alpha information set. In Panels B and C, which intervene only on Alpha processing, the No-Alpha accuracy is exactly invariant within each symbol and panel: 45.0% for both normalisation sweeps, 50.0% for the FPT weight sweep, and 45.0% for the VNM weight sweep.

The stored alpha-enabled branch varies with configuration. These sweeps do not validate historical branch provenance or establish alpha information value: snapshot dates differ between experiments, repeats are absent, and the control may be weak. Their arithmetic is retained for traceability. The new study is frozen at one configuration and does not retrospectively authenticate these sweeps or establish configuration robustness.

6.8. Unverified Legacy Alpha–Sentiment Ablation

Table 17 retains the legacy four-way summaries for traceability only. Identical prediction sequences across differently labelled FPT branches and missing request archives leave provenance unresolved. The numerical gaps in this table therefore cannot be called independent alpha or sentiment contributions. Its final column is the stored improvement over the best single component, not the factorial interaction.

No component-information conclusion is drawn from these unverified summaries. Their presence preserves the audit trail. The newly archived common-data factorial/placebo panel is reported separately above; its mixed contrasts do not validate the legacy summaries or establish component information value.

Table 13: Paired benchmark over nine equities ($N = 20$ per symbol, $W = 45$, $S = 3$, $L = 3$). Panel A reports directional inference; Panel B reports compounded long-only account outcomes.

Panel A: Directional accuracy and paired inference

Symbol	Full (%)	No-α (%)	Lift (pp)	95% CI (pp)	McNemar p	N
BHN	55.0	45.0	+10.0	[+0.0, +15.0]	0.6875	20
CMG	75.0	70.0	+5.0	[+0.0, +5.0]	1.0000	20
FPT	65.0	45.0	+20.0	[−5.0, +35.0]	0.2188	20
HVN	70.0	50.0	+20.0	[+0.0, +35.0]	0.2188	20
MBB	70.0	45.0	+25.0	[+0.0, +55.0]	0.1250	20
MWG	70.0	50.0	+20.0	[+0.0, +40.0]	0.1250	20
VCB	65.0	50.0	+15.0	[+5.0, +35.0]	0.2500	20
VJC	60.0	60.0	+0.0	[−10.0, +10.0]	1.0000	20
VNM	55.0	45.0	+10.0	[−15.0, +45.0]	0.6250	20
Mean / total N	**65.0**	**51.1**	**+13.9**	–	–	**180**

Panel B: Compounded account metrics

Symbol	Return F (%)	Return N (%)	Sharpe F	Sharpe N	MDD F (%)	MDD N (%)
BHN	-4.20	-5.76	-3.63	-2.64	4.80	8.31
CMG	-2.66	-5.60	-3.64	-5.28	2.66	5.60
FPT	-1.56	-10.37	-0.86	-6.41	6.30	10.37
HVN	+2.89	-15.64	+1.19	-4.27	5.27	20.06
MBB	+5.21	-7.78	+2.36	-4.22	3.50	9.89
MWG	+0.39	-6.42	+0.41	-3.93	1.92	8.17
VCB	-8.34	-10.49	-5.98	-7.78	8.92	10.49
VJC	-6.05	-6.26	-5.22	-5.38	6.05	6.26
VNM	-0.85	-7.51	-0.41	-4.91	3.40	7.51
Mean	**-1.69**	**-8.43**	**-1.75**	**-4.98**	**4.76**	**9.63**

F and N denote Full and No-Alpha. The 95% intervals use a moving-block bootstrap (block length 5; 1,000 replications). The two-sided cross-symbol Wilcoxon result is $p = 0.0078125$ (nine symbol-level lifts; median lift = 15.0 pp). Returns are compounded from VND 50,000,000 independent accounts under Eq. (34); SHORT forecasts remain in cash.

7. Limitations & Future Work

7.1. Limitations

L1. Statistical scale and economic interpretation.. The legacy benchmark contains only 20 observations per symbol. Common dates and market shocks invalidate interpreting nine symbol-level gaps as nine independent market replications, and the stored Wilcoxon result is not used to establish a general significant improvement. The tickers are not a probability sample. Class imbalance, snapshot revisions and incomplete input provenance further constrain inference even for this fixed universe.

The new study extends historical coverage to six blocks, but contains only twelve distinct decision dates and 36 distinct target-session dates in a purposively selected nine-symbol universe. These dates, the symbol accounts and the repeated calls are not additional independent market replications. BHN covers only three recent blocks; cohort sensitivities preserve that missingness. Joint block bootstrap

Table 14: Exploratory numerical baselines on the newly retrieved snapshot: not a head-to-head comparison with legacy Full.

Variant	Acc. (%)	BA (%)	MCC	Mean return (%)	Mean daily MDD (%)	Held sessions (%)
Constant DOWN / CASH	68.89	50.00	NA	0.00	0.00	0.00
Always-LONG cycles	31.11	50.00	NA	-17.33	21.13	99.15
Continuous hold (price only)	NA	NA	NA	-5.53	14.74	100.00
Factor-only vote	61.67	62.38	0.230	-5.07	11.93	46.75
Logistic: price	66.11	52.88	0.077	-1.15	4.22	13.80
Logistic: price + factors + S/M	61.67	51.12	0.025	-2.83	7.38	21.62
Momentum (20)	50.00	46.08	-0.074	-8.55	10.50	40.60
Train majority	68.89	50.00	NA	0.00	0.00	0.00
Train-rate random control	50.56	47.47	-0.047	-10.03	12.06	42.44

All cycle classifiers use the same 180 new-snapshot cutoffs and net targets. Predictive metrics are pooled; economic columns average independent symbol accounts. S/M denotes verified sentiment and missingness; historical scorer provenance is unavailable, so sentiment scores are missing. The train-rate control uses training momentum exposure, not Full exposure. Continuous hold has one buy and one sell per symbol; price adjustments and dividends are not independently certified. No Full row is included because 74 cycles differ at entry and/or exit from legacy, including two changed labels.

Table 15: Training support and snapshot comparability audit.

Symbol	Train	Validation	Test	Different endpoints	Label changes
BHN	175	34	20	20	2
CMG	399	36	20	1	0
FPT	399	36	20	20	0
HVN	399	36	20	1	0
MBB	399	36	20	1	0
MWG	399	36	20	20	0
VCB	399	36	20	9	0
VJC	399	36	20	0	0
VNM	399	36	20	2	0

Each label exit precedes the next split's first decision. BHN uses a valid suffix beginning 14 September 2022 after an unverified OHLC contradiction; raw data are preserved. All 20 test dates are retained for every symbol. Matching aggregate class counts does not establish matching labels or snapshots.

intervals and preselected repeated panels are descriptive. With six or three historical clusters, neither power nor inter-block independence is established. The Full–Base gap changes sign when individual blocks are omitted. Volatility/turnover strata are causal calculations added after inference and cannot support a confirmatory regime strategy. Historical prices may be represented in either hosted model's training corpus: all target labels precede protocol freeze, and no served weight digest or authenticated training exclusion is available. Model release date alone does not remove that concern. Full's new BA of 48.31% and negative mean return do not establish a benefit over naive or numerical controls.

Directional skill does not directly imply profitability. In the legacy run, Full improves mean accuracy by 13.9 pp over Base and has a less negative mean account return, yet remains below CASH. The new study likewise has negative Full mean net return. Because SHORT predictions map to CASH, correct bearish forecasts generate no trading return. Relative loss reduction does not establish deployable profitability.

Table 16: Robustness sweep on FPT and VNM ($N = 20$ per row). Returns are compounded account returns.

Symbol	Panel	Scenario	Full (%)	No-α (%)	Lift (pp)	Return F (%)	Return N (%)
FPT	Window	$W = 30$	55.0	40.0	+15.0	-2.63	-10.27
FPT	Window	$W = 45$	65.0	50.0	+15.0	-2.21	-6.86
FPT	Window	$W = 60$	40.0	45.0	-5.0	-8.04	-4.38
VNM	Window	$W = 30$	40.0	40.0	+0.0	-7.81	-7.51
VNM	Window	$W = 45$	55.0	40.0	+15.0	-3.59	-7.06
VNM	Window	$W = 60$	65.0	55.0	+10.0	+5.20	-0.84
FPT	Normalisation	`zscore_tanh`	50.0	45.0	+5.0	-4.72	-7.38
FPT	Normalisation	`minmax`	55.0	45.0	+10.0	-5.63	-7.38
FPT	Normalisation	`rank`	60.0	45.0	+15.0	-4.36	-7.38
VNM	Normalisation	`zscore_tanh`	60.0	45.0	+15.0	+2.70	-2.22
VNM	Normalisation	`minmax`	65.0	45.0	+20.0	+3.89	-2.22
VNM	Normalisation	`rank`	55.0	45.0	+10.0	-2.01	-2.22
FPT	Weighting	`default`	60.0	50.0	+10.0	-1.99	-7.85
FPT	Weighting	`equal_weight`	60.0	50.0	+10.0	-0.36	-7.85
FPT	Weighting	`ic_heavy`	65.0	50.0	+15.0	-2.21	-7.85
VNM	Weighting	`default`	65.0	45.0	+20.0	+6.17	-5.19
VNM	Weighting	`equal_weight`	50.0	45.0	+5.0	-4.57	-5.19
VNM	Weighting	`ic_heavy`	60.0	45.0	+15.0	+2.70	-5.19

Table 17: Unverified legacy four-way summaries ($N = 20$ per variant and symbol); not confirmatory attribution evidence.

Panel A: Directional decomposition

Symbol	Full	Alpha-only	Sent.-only	Baseline	$\Delta\alpha$	ΔSent.	Best-single gap
FPT	55.0%	50.0%	45.0%	45.0%	+5.0 pp	+0.0 pp	+5.0 pp
VNM	60.0%	50.0%	50.0%	45.0%	+5.0 pp	+5.0 pp	+10.0 pp
VCB	45.0%	55.0%	40.0%	40.0%	+15.0 pp	+0.0 pp	-10.0 pp
Mean	**53.33%**	**51.67%**	**45.00%**	**43.33%**	**+8.33 pp**	**+1.67 pp**	**+1.67 pp**

Panel B: Compounded account return

Symbol	Full	Alpha-only	Sentiment-only	Baseline
FPT	-4.61%	-7.85%	-7.38%	-6.09%
VNM	-4.03%	+2.26%	-5.86%	-3.96%
VCB	-7.93%	-2.75%	-9.23%	-11.34%

L2. Simplified execution model.. The fixed-cycle ledger aligns labels, alpha targets, and account returns on the same next-open-to-target-close net return and includes two-leg fees and slippage. However, these costs remain benchmark assumptions. The simulator omits dynamic spreads, taxes, order-book depth, partial fills, market impact, position limits, and cross-symbol capital competition. It also assumes each $S = L = 3$ cycle is closed before the next entry. A venue-calibrated event-driven simulator is therefore required before live deployment.

New daily ledgers additionally assume fractional shares and immediate cash reuse. Closing a share position does not establish settlement of sale proceeds; pending cash or funded advances have

not been validated. Adjustment/dividend provenance is uncertain, so continuous hold is not certified total return.

L3. Alpha selection and sensitivity.. Selecting five factors from 85 candidates introduces multiple-testing and winner's-curse risk. Although point-in-time cutoffs prevent future-bar leakage, the same trailing sample is used to estimate IC sign, compute oriented Accuracy and Sharpe, and rank the factors. Selecting large composite scores across many candidates can therefore produce optimistic diagnostics. No nested cross-validation, Deflated Sharpe Ratio, multiplicity correction, or sign-stability criterion was applied. The benchmark evaluates the complete adaptive policy at later cutoffs, but its per-factor statistics should not be interpreted as unbiased factor discoveries.

The 80 WorldQuant-derived entries are heuristic single-asset adaptations rather than exact implementations of the original cross-sectional formulas. Their stability was not evaluated through IC decay, lagged signal autocorrelation, or turnover. These diagnostics are needed to determine whether apparent predictiveness survives to the $L = 3$ horizon and remains economically meaningful after trading costs.

L4. Hosted-model and visual reliability.. Configured temperature zero is mapped to 10^{-8} by the audited SDK; fixed prompts and that near-zero client setting do not guarantee deterministic hosted outputs. Sharing upstream reports controls those inputs within a pair, not randomness of separate Decision calls. The three-execution audit reports actual conditional dispersion rather than assuming it vanishes. Provider routing, timing and unverified served weights remain confounds. The Decision and upstream panels have different supports and only one overlapping unit; they cannot identify separate variance components without a crossed design. Three runs and three blocks, with substantial undefined resamples, do not justify precise or calibrated confidence claims or generalisation to all 102 units. Nominal legacy defaults are not authenticated matched repeats.

The Pattern and Trend agents use deterministic OHLCV rules to detect obvious inconsistencies in visual conclusions. These rules check output consistency only and leave the VLM's actual use of the cited candle, wick, breakout, or trendline unverified. Prior work reports strong trend sensitivity and persistent spatial-reasoning limitations in financial chart understanding [11, 10]. Establishing causal visual grounding would require matched counterfactual charts and human-labelled local evidence, neither of which is included in this study.

Factor-value controls and attribution limits. The original compressed prompts omitted the numerical factor rows. The separate completed 270-decision repair verifies their delivery and compares a joint past rank-vector placebo under the same report structure. It does not preserve named-factor covariance, exactly match token lengths, isolate one scalar from its skill metadata or match LONG exposure. AS–P has a positive pointwise BA interval but a zero-spanning seven-contrast envelope; all branches retain UP recall of 16.67% and negative mean net return. Three executions/blocks, coverage-selected historical units, shared rules and provider timing prevent a general information-value or calibrated significance conclusion. The original and repaired results remain separate.

L5. Sentiment coverage and attribution.. Cutoff filtering is an intended software control, not independent verification of archived inputs. Historical caches lack scorer provenance and are treated as unavailable in the verified-score numerical comparator, rather than as neutral sentiment. The three-symbol legacy ablation has unresolved branch provenance and does not establish independent sentiment effects or interactions. New provenance fields cannot retroactively identify the scorer or authenticate the missing historical requests.

The new bank pins a local scorer and archives its inputs and scores, but does not independently certify publication/content availability or Vietnamese financial-domain validity. Its coverage-selected 18-unit panel has only five dates and excludes CMG; the nine-unit upstream panel answers a different support-specific question. Mixed placebo contrasts do not support isolated alpha information value. Neither panel closes the separate source, domain-validation or component-reliability requirements.

7.2. Future Work

F1. Larger and longer evaluation.. The most important extension is a preregistered study over a broader universe such as the VN100 and multiple market regimes. More test points would increase per-symbol power, support sector-level analysis, and permit out-of-time validation rather than repeated analysis of the same recent period.

F2. Stronger baselines and multiplicity control.. The new study already includes train-fitted majority, momentum, two logistic feature sets and factor voting. Further tree-based or sequence controls and a time-ordered nested design would be useful if stronger component-selection claims are pursued. Normalisation, IC orientation and top-five selection should then be repeated wholly inside each inner training fold before scoring the frozen bundle on an untouched outer interval [17]. Multiplicity controls and selection/sign/rank stability are still needed before interpreting selected-factor diagnostics as factor discoveries.

F3. Execution and portfolio realism.. A portfolio simulator should model lot sizes, taxes, dynamic spreads, liquidity limits, partial fills, and market impact, and should allocate capital jointly across symbols. Confidence gates and risk-based sizing can then be assessed without conflating forecasting accuracy with all-in binary allocation.

F4. Component reliability.. Broader repeated hosted-model studies should extend the two limited panels and archive provider revision metadata whenever available. The new local ViSoBERT scorer is pinned; backend-equivalence tests would be required before pooling it with Dedicated Endpoint outputs. Vision faithfulness needs matched local-pattern/global-trend interventions and human annotations. Sentiment also needs a Vietnamese financial-domain test set. Unverified legacy interactions are not used as evidence to choose a new conflict-resolution mechanism after observing outcomes.

F5. Per-factor translation audit.. Each registered adaptation should be profiled by horizon-specific IC decay, autocorrelation at several lags, sign persistence, exposure turnover, and net performance after the same two-leg costs used by the account simulator. Diagnostics should be stratified by symbol and market regime and compared with simple price/volume baselines. Factors whose orientation changes frequently or whose apparent IC disappears before $L = 3$ should be excluded by a preregistered stability rule rather than retained through ad hoc interpretation.

7.3. Broader Impact

Localized financial AI can broaden research beyond English-language, highly liquid markets, but automated use in a shallow market may amplify noise and herding. The framework should therefore be treated as decision-support research rather than trading advice. Reproducible releases must preserve model and data provenance while respecting news copyright, provider terms, and privacy obligations.

8. Conclusion

The new frozen common-snapshot evaluation completes all 474 planned research decisions, including 102 Base–Full pairs across nine Vietnamese equities, six historical blocks and twelve decision dates. Full achieves 62.75% raw accuracy, below constant CASH at 75.49%; its balanced accuracy is 48.31% and MCC is −0.035, versus 55.61% balanced accuracy for price-only logistic regression. Its mean net return over 51 independent symbol–block accounts is −0.88%, below CASH at zero and price logistic at −0.68%. The evidence therefore does not establish predictive superiority or profitability. The Full–Base balanced-accuracy difference is only 0.55 percentage points, with a descriptive six-block interval spanning zero.

Preselected three-repeat decision/factorial/placebo and upstream panels quantify variation only on their limited supports; their heterogeneous results do not establish isolated factor or sentiment value. Train-fitted majority, momentum, two logistic feature sets, factor voting and continuous holding provide stronger context than the weaker Base reports control. Both-class metrics, explicit common support and daily ledgers make the negative comparison inspectable. The new experiment remains historical and exploratory: provider revisions and model contamination are unverified, news availability and financial-domain sentiment validity are uncertain, and fractional immediate-cash ledgers do not certify settlement or deployability. The separate 180-decision legacy audit is preserved, with its 65.0% Full accuracy below 68.9% CASH; it is not pooled with the new study because authenticated historical inputs remain unavailable. A broader independent evaluation and validated execution/input provenance would be needed to support stronger claims.

The temporal audit documents 33 rising, 35 falling and 34 sideways individual price contexts, with train-calibrated descriptive volatility and turnover strata. It preserves calendar missingness and cross-sectional shocks in resampling. Twelve historical decision dates and six blocks provide limited evidence, not independent market replications or a contamination-free prospective test.

The matched Decision-repeat panel has a Full–Base BA difference of 5.56 percentage points with sample SD 14.63 points across three executions. Its sign varies, and conditional few-block intervals are not calibrated confidence guarantees. The separately supported upstream panel cannot identify a variance decomposition. These distributions strengthen the case for retaining negative results and avoiding best-run or deterministic-hosted-model claims.

Actual-prompt inspection found that the original alpha distiller omitted the five factor-value rows. A separately frozen 270-decision repaired panel now verifies complete delivery and genuine fresh factorial/placebo calls. Its Full–placebo BA difference averages 6.94 points, but the seven-contrast descriptive envelope spans zero; UP recall stays at 16.67% across branches, and Full BA 44.44% and mean net return −0.670% remain below CASH references. The matched rank-vector control therefore addresses prompt content more directly without establishing general alpha information value, independent sentiment benefit, positive synergy or profitability. All old predictions and their distinct uncertainty statements are retained.

CRediT authorship contribution statement

Van-Tang Nguyen: Conceptualization (C); Software (So); Validation (Va); Investigation (I); Original draft preparation (O); Project administration (P); Funding acquisition (Fu).

Hoang-Dung Phan: Methodology (M); Software (So); Formal analysis (Fo); Investigation (I); Resources (R); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Visualization (Vi); Supervision (Su).

Thanh-Tam Do Thi: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Supervision (Su).

Phuong-Thai Nguyen: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Visualization (Vi); Supervision (Su).

Hong-Viet Tran: Conceptualization (C); Methodology (M); Software (So); Validation (Va); Formal analysis (Fo); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Project administration (P).

CONFLICT OF INTEREST STATEMENT (mandatory)

Authors state no conflict of interest.

CODE AVAILABILITY

The source code and demonstration implementation of the proposed framework are available in the project repository.

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Appendix A. New Common-Snapshot Baseline Details

These tables belong only to the new 102-unit study. Percentages and support refer to net-cycle outcomes; NA denotes undefined or inapplicable metrics. The equal-symbol means below differ from pooled weighting because BHN has six units while each other symbol has twelve. Every classifier's full confusion and both-class metrics, including groups with absent classes, are retained in the accompanying CSV/JSON release.

Table A.18: Equal-symbol macro means, with defined denominators out of nine symbols.

Variant	Acc.	BA	Defined BA	MCC	Defined MCC	LONG
Full (AS)	62.96	45.96	8	-0.106	8	22.22
Base reports (B)	58.33	50.07	8	-0.020	8	32.41
Constant CASH	75.93	50.00	8	NA	0	0.00
Always-LONG cycles	24.07	50.00	8	NA	0	100.00
Factor vote	50.00	46.86	8	-0.042	8	50.00
Logistic market inputs	62.04	56.55	8	0.082	8	32.41
Logistic price	68.52	59.00	8	0.174	7	24.07
Momentum (20)	49.07	50.67	8	-0.008	8	49.07
Train majority	75.93	50.00	8	NA	0	0.00

Percentages except MCC. Means omit undefined symbol metrics; defined counts are shown. BHN has six observations in three blocks; the other eight symbols have twelve in six blocks. This differs from pooled weighting.

Table A.19: Every baseline by symbol on common support; percentages except MCC. (Part 1 of 3.)

Symbol	Variant	N	UP	Acc.	BA	MCC	LONG	Return
BHN	Full (AS)	6	1	66.67	40.00	-0.200	16.67	-1.23
BHN	Base reports (B)	6	1	66.67	80.00	0.447	50.00	-1.53
BHN	Constant CASH	6	1	83.33	50.00	NA	0.00	0.00
BHN	Always-LONG cycles	6	1	16.67	50.00	NA	100.00	-3.76
BHN	Train majority	6	1	83.33	50.00	NA	0.00	0.00
BHN	Momentum (20)	6	1	50.00	70.00	0.316	66.67	-2.64
BHN	Logistic price	6	1	83.33	90.00	0.632	33.33	-1.01
BHN	Logistic market inputs	6	1	83.33	90.00	0.632	33.33	-1.01
BHN	Factor vote	6	1	16.67	10.00	-0.632	66.67	-2.64
BHN	Continuous hold	NA	NA	NA	NA	NA	NA	-2.11
CMG	Full (AS)	12	3	66.67	44.44	-0.174	8.33	-0.43
CMG	Base reports (B)	12	3	75.00	61.11	0.258	16.67	-0.03
CMG	Constant CASH	12	3	75.00	50.00	NA	0.00	0.00
CMG	Always-LONG cycles	12	3	25.00	50.00	NA	100.00	-3.20
CMG	Train majority	12	3	75.00	50.00	NA	0.00	0.00
CMG	Momentum (20)	12	3	75.00	61.11	0.258	16.67	-0.16
CMG	Logistic price	12	3	66.67	66.67	0.293	41.67	-2.12
CMG	Logistic market inputs	12	3	66.67	66.67	0.293	41.67	-0.74
CMG	Factor vote	12	3	66.67	66.67	0.293	41.67	-1.26
CMG	Continuous hold	NA	NA	NA	NA	NA	NA	-13.30
FPT	Full (AS)	12	2	58.33	35.00	-0.258	25.00	-1.03
FPT	Base reports (B)	12	2	50.00	30.00	-0.316	33.33	-1.61
FPT	Constant CASH	12	2	83.33	50.00	NA	0.00	0.00
FPT	Always-LONG cycles	12	2	16.67	50.00	NA	100.00	-2.87
FPT	Train majority	12	2	83.33	50.00	NA	0.00	0.00
FPT	Momentum (20)	12	2	50.00	50.00	0.000	50.00	-1.01
FPT	Logistic price	12	2	58.33	55.00	0.076	41.67	0.21
FPT	Logistic market inputs	12	2	50.00	30.00	-0.316	33.33	-2.19
FPT	Factor vote	12	2	41.67	45.00	-0.076	58.33	-2.38
FPT	Continuous hold	NA	NA	NA	NA	NA	NA	-0.51

NA denotes undefined or inapplicable; all variants use common support.

Table A.20: Every baseline by symbol on common support; percentages except MCC. (Part 2 of 3.)

Symbol	Variant	N	UP	Acc.	BA	MCC	LONG	Return
HVN	Full (AS)	12	2	58.33	55.00	0.076	41.67	-2.49
HVN	Base reports (B)	12	2	58.33	55.00	0.076	41.67	-2.49
HVN	Constant CASH	12	2	83.33	50.00	NA	0.00	0.00
HVN	Always-LONG cycles	12	2	16.67	50.00	NA	100.00	-3.95
HVN	Train majority	12	2	83.33	50.00	NA	0.00	0.00
HVN	Momentum (20)	12	2	58.33	75.00	0.378	58.33	-1.99
HVN	Logistic price	12	2	75.00	45.00	-0.135	8.33	-0.78
HVN	Logistic market inputs	12	2	75.00	85.00	0.529	41.67	-0.39
HVN	Factor vote	12	2	50.00	50.00	0.000	50.00	-1.15
HVN	Continuous hold	NA	NA	NA	NA	NA	NA	-15.91
MBB	Full (AS)	12	5	58.33	55.71	0.120	33.33	-0.05
MBB	Base reports (B)	12	5	58.33	55.71	0.120	33.33	-0.05
MBB	Constant CASH	12	5	58.33	50.00	NA	0.00	0.00
MBB	Always-LONG cycles	12	5	41.67	50.00	NA	100.00	-1.88
MBB	Train majority	12	5	58.33	50.00	NA	0.00	0.00
MBB	Momentum (20)	12	5	25.00	24.29	-0.507	50.00	-2.08
MBB	Logistic price	12	5	58.33	52.86	0.076	16.67	-0.96
MBB	Logistic market inputs	12	5	58.33	55.71	0.120	33.33	-0.50
MBB	Factor vote	12	5	66.67	65.71	0.314	41.67	-0.20
MBB	Continuous hold	NA	NA	NA	NA	NA	NA	0.83
MWG	Full (AS)	12	4	58.33	56.25	0.120	41.67	-0.46
MWG	Base reports (B)	12	4	41.67	43.75	-0.120	58.33	-1.03
MWG	Constant CASH	12	4	66.67	50.00	NA	0.00	0.00
MWG	Always-LONG cycles	12	4	33.33	50.00	NA	100.00	-3.12
MWG	Train majority	12	4	66.67	50.00	NA	0.00	0.00
MWG	Momentum (20)	12	4	33.33	37.50	-0.250	66.67	-2.29
MWG	Logistic price	12	4	58.33	56.25	0.120	41.67	-1.76
MWG	Logistic market inputs	12	4	58.33	56.25	0.120	41.67	-2.05
MWG	Factor vote	12	4	41.67	37.50	-0.239	41.67	-2.36
MWG	Continuous hold	NA	NA	NA	NA	NA	NA	2.27

NA denotes undefined or inapplicable; all variants use common support.

Table A.21: Every baseline by symbol on common support; percentages except MCC. (Part 3 of 3.)

Symbol	Variant	N	UP	Acc.	BA	MCC	LONG	Return
VCB	Full (AS)	12	4	50.00	37.50	-0.316	16.67	-0.93
VCB	Base reports (B)	12	4	41.67	31.25	-0.408	25.00	-1.04
VCB	Constant CASH	12	4	66.67	50.00	NA	0.00	0.00
VCB	Always-LONG cycles	12	4	33.33	50.00	NA	100.00	-2.31
VCB	Train majority	12	4	66.67	50.00	NA	0.00	0.00
VCB	Momentum (20)	12	4	33.33	25.00	-0.500	33.33	-2.28
VCB	Logistic price	12	4	66.67	50.00	NA	0.00	0.00
VCB	Logistic market inputs	12	4	50.00	37.50	-0.316	16.67	-0.93
VCB	Factor vote	12	4	50.00	56.25	0.125	66.67	-0.89
VCB	Continuous hold	NA	NA	NA	NA	NA	NA	-0.85
VJC	Full (AS)	12	4	58.33	43.75	-0.213	8.33	-0.77
VJC	Base reports (B)	12	4	58.33	43.75	-0.213	8.33	-0.77
VJC	Constant CASH	12	4	66.67	50.00	NA	0.00	0.00
VJC	Always-LONG cycles	12	4	33.33	50.00	NA	100.00	-2.00
VJC	Train majority	12	4	66.67	50.00	NA	0.00	0.00
VJC	Momentum (20)	12	4	58.33	62.50	0.239	58.33	-1.03
VJC	Logistic price	12	4	66.67	56.25	0.158	16.67	0.46
VJC	Logistic market inputs	12	4	41.67	31.25	-0.408	25.00	-1.23
VJC	Factor vote	12	4	41.67	43.75	-0.120	58.33	-1.68
VJC	Continuous hold	NA	NA	NA	NA	NA	NA	8.33
VNM	Full (AS)	12	0	91.67	NA	NA	8.33	-0.74
VNM	Base reports (B)	12	0	75.00	NA	NA	25.00	-0.99
VNM	Constant CASH	12	0	100.00	NA	NA	0.00	0.00
VNM	Always-LONG cycles	12	0	0.00	NA	NA	100.00	-4.30
VNM	Train majority	12	0	100.00	NA	NA	0.00	0.00
VNM	Momentum (20)	12	0	58.33	NA	NA	41.67	-1.74
VNM	Logistic price	12	0	83.33	NA	NA	16.67	-0.36
VNM	Logistic market inputs	12	0	75.00	NA	NA	25.00	-0.87
VNM	Factor vote	12	0	75.00	NA	NA	25.00	-1.15
VNM	Continuous hold	NA	NA	NA	NA	NA	NA	2.12

NA denotes undefined or inapplicable; all variants use common support.

Return in the symbol table is the mean of independent two-cycle symbol–block accounts (three accounts for BHN and six for each other symbol). Continuous hold uses one buy and sell per account and has no cycle-classification scores. This is a price-series comparator under the same assumed costs, not certified total return. Per-account daily-close drawdown, turnover, fees, slippage, held-session fraction and close exposure are provided in the economic CSV and daily ledgers.

Table A.22: Full, Base, CASH and price logistic by historical block and causal regime. (Part 1 of 2.)

Stratum	Variant	N	UP	Acc.	BA	MCC
b2023_h2	Full (AS)	16	3	68.75	42.31	-0.182
b2023_h2	Base reports (B)	16	3	43.75	26.92	-0.372
b2023_h2	Constant CASH	16	3	81.25	50.00	NA
b2023_h2	Logistic price	16	3	62.50	64.10	0.222
b2024_h2	Full (AS)	16	7	43.75	42.06	-0.163
b2024_h2	Base reports (B)	16	7	50.00	49.21	-0.016
b2024_h2	Constant CASH	16	7	56.25	50.00	NA
b2024_h2	Logistic price	16	7	50.00	46.03	-0.101
b2025_h1	Full (AS)	16	0	87.50	NA	NA
b2025_h1	Base reports (B)	16	0	87.50	NA	NA
b2025_h1	Constant CASH	16	0	100.00	NA	NA
b2025_h1	Logistic price	16	0	81.25	NA	NA
f2025_q4	Full (AS)	18	7	66.67	59.74	0.255
f2025_q4	Base reports (B)	18	7	61.11	55.19	0.122
f2025_q4	Constant CASH	18	7	61.11	50.00	NA
f2025_q4	Logistic price	18	7	61.11	57.79	0.161
f2026_q2	Full (AS)	18	6	50.00	37.50	-0.316
f2026_q2	Base reports (B)	18	6	55.56	45.83	-0.094
f2026_q2	Constant CASH	18	6	66.67	50.00	NA
f2026_q2	Logistic price	18	6	72.22	62.50	0.316
f2026_q3	Full (AS)	18	2	61.11	56.25	0.081
f2026_q3	Base reports (B)	18	2	50.00	50.00	0.000
f2026_q3	Constant CASH	18	2	88.89	50.00	NA
f2026_q3	Logistic price	18	2	77.78	43.75	-0.125

NA denotes undefined or inapplicable; all variants use common support.

Table A.23: Full, Base, CASH and price logistic by historical block and causal regime. (Part 2 of 2.)

Stratum	Variant	N	UP	Acc.	BA	MCC
bear	Full (AS)	35	5	71.43	41.67	-0.167
bear	Base reports (B)	35	5	71.43	41.67	-0.167
bear	Constant CASH	35	5	85.71	50.00	NA
bear	Logistic price	35	5	74.29	60.00	0.167
bull	Full (AS)	33	9	57.58	50.00	0.000
bull	Base reports (B)	33	9	48.48	47.22	-0.050
bull	Constant CASH	33	9	72.73	50.00	NA
bull	Logistic price	33	9	57.58	46.53	-0.069
sideways	Full (AS)	34	11	58.82	48.22	-0.041
sideways	Base reports (B)	34	11	52.94	46.25	-0.075
sideways	Constant CASH	34	11	67.65	50.00	NA
sideways	Logistic price	34	11	70.59	61.66	0.270

NA denotes undefined or inapplicable; all variants use common support.

Regime is causal per symbol using the decision-close/SMA60 gap. The machine-readable classification release additionally contains all nine classifiers by block/regime and the common eight-symbol/six-block and nine-symbol/three-recent-block cohorts. Blocks and dates are shared across symbols, so neither symbol counts nor repeated decisions create additional independent market replications.

Appendix A.1. Temporal Coverage and Regime Audit

The R2 audit reuses the fixed 102-unit prediction bank. Exact train, validation, test and label-maturity boundaries for every symbol–block scope are provided in `folds.csv`. The data manifest records source, quality and universe limits; the calendar-panel CSV retains every missing cell. No missing prediction is replaced with CASH and no unit is added after observing an outcome.

Table A.24: Temporal support of the fixed R1 prediction bank.

Block	Dates (MM-DD)	Symbols	N	UP/DOWN	Bull/Bear/Side	LONG AS	LONG B
b2023_h2	07-03/09-22	8	16	3/13	9/2/5	2	6
b2024_h2	07-01/09-20	8	16	7/9	8/4/4	6	7
b2025_h1	01-02/03-31	8	16	0/16	5/6/5	2	2
f2025_q4	10-01/12-19	9	18	7/11	5/7/6	3	4
f2026_q2	04-01/06-24	9	18	6/12	1/13/4	3	4
f2026_q3	07-01/09-23	9	18	2/16	5/3/10	7	9

Each block has two decision dates and six distinct target-session dates. The first three omit BHN for the frozen valid-suffix/warm-up rule. Totals: 102 units, 12 decision dates, 36 target-session calendar dates; none is asserted statistically independent.

Table A.25: Audit of the frozen chronological outer folds and inner validation.

Block	Symbol scopes	Train range	Validation range	Warm-up	Purge	Refit stats
b2023_h2	8	70–70	19–19	600	PASS	PASS
b2024_h2	8	153–153	19–19	600	PASS	PASS
b2025_h1	8	194–194	21–21	600	PASS	PASS
f2025_q4	9	32–256	20–20	600	PASS	PASS
f2026_q2	9	75–299	18–18	600	PASS	PASS
f2026_q3	9	94–318	20–20	600	PASS	PASS

Warm-up counts available observed sessions at the first train cutoff. All 51 scopes have label exits strictly before the next split. Stored imputation/scaling statistics reproduce from train plus matured validation; classifiers are not refitted. Exact split dates/counts are in folds.csv.

Table A.26: Causal context strata on the same predictions (percentages except counts).

Dimension	Stratum	N	Dates	UP	BA AS	BA B	BA price	LONG AS
Trend	bear	35	12	5	41.67	41.67	60.00	14.29
Trend	bull	33	11	9	50.00	47.22	46.53	33.33
Trend	sideways	34	11	11	48.22	46.25	61.66	20.59
Volatility	high	33	10	13	46.54	45.38	56.54	27.27
Volatility	low	69	11	12	47.81	47.59	58.55	20.29
Turnover proxy	high	80	12	20	46.67	45.83	52.50	25.00
Turnover proxy	low	22	7	5	54.12	55.29	68.24	13.64

Trend uses the frozen close/SMA60 rule. Volatility and turnover-proxy high/low thresholds are symbol/fold train-only medians; these two dimensions are post-inference descriptive additions. Dates overlap across strata. Turnover proxy is not order-book liquidity; no regime-dependent strategy is fitted.

Table A.27: Coverage-cohort sensitivity with synchronized whole-block resampling.

Cohort	N	Blocks	Dates	BA AS	BA B	Delta (pp)	95% interval (pp)
All available	102	6	12	48.31	47.77	0.55	[-5.91, 8.00]
Eight symbols	96	6	12	48.61	46.53	2.08	[-4.10, 8.67]
Nine recent	54	3	6	47.18	46.67	0.51	[-8.33, 6.25]

10,000 descriptive resamples, seed 20261102. Every selected block retains its full cross-section, both dates, all variants and missing eligibility mask. Cohorts overlap; three or six blocks are too few for a precision or general-superiority claim.

The machine-readable release additionally provides class support and both classes' metrics for all nine classifiers within each context stratum, target-session exposure verified against daily ledgers, all paired bootstrap contrasts for three overlapping coverage cohorts, and leave-one-block-out sensitivities. Volatility and turnover medians are fitted only on train; their use here is explicitly post-inference description. The nine-symbol universe remains purposive and potentially survivor-biased. Distinct dates are counts of observations, not a claim about the effective independent sample size. `contamination.json` records zero units after protocol freeze and no authenticated served-model training cutoff.

Appendix A.2. Repeated Execution Identity and Distributions

The R3 release links every panel decision to its actual request ID, prompt/asset/upstream hashes, timestamp, provider key slot and execution epoch. It retains all 756 attempts (including 22 errors),

132 panel-related node records and 322 unique panel decision executions. Fixed-upstream Decision repeats have identical actual prompts within all 90 unit–variant groups. The one overlapping unit shares its r0 B/AS pair by design, while every distinct repeat/realization has its own successful response. The audit also records actual call order and unknown legacy input/prompt/model provenance.

Metrics for all three executions, paired contrasts, mean/SD/median/range, conditional percentile intervals and defined/undefined resample counts are available for accuracy, BA, MCC, macro-F1, LONG frequency and mean account return. The daily ledger CSV contains 249 panel account executions; its 13-account and nine-account supports differ from the broad 51-account study. No $3N$-market pooled accuracy, best-run result or ensemble is used.

Table A.28: Accuracy, MCC and LONG frequency across the same three executions.

Panel	Variant	N	Acc. mean	Acc. SD (pp)	MCC mean	MCC SD	LONG mean
decision	B	18	27.78	5.56	-0.473	0.079	50.00
decision	A	18	35.19	8.49	-0.372	0.112	42.59
decision	S	18	31.48	3.21	-0.421	0.044	46.30
decision	AS	18	37.04	11.56	-0.378	0.195	37.04
decision	P	18	38.89	5.56	-0.323	0.073	38.89
upstream	B	9	85.19	6.42	0.703	0.091	37.04
upstream	AS	9	96.30	6.42	0.919	0.141	25.93

Percentages except MCC. Per-execution scores and mean/SD/median/range/conditional intervals for every metric are retained in the machine-readable report. These are panel-only distributions, not 102-unit one-pass uncertainty.

Table A.29: Selected-panel balanced accuracy across three actual executions (percent).

Panel	Variant	N	Mean	SD (pp)	Median	Range	Descriptive interval
decision	B	18	25.00	4.17	25.00	[20.83, 29.17]	[16.67, 37.82]
decision	A	18	30.56	6.36	29.17	[25.00, 37.50]	[16.67, 50.64]
decision	S	18	27.78	2.41	29.17	[25.00, 29.17]	[14.29, 47.22]
decision	AS	18	30.56	10.49	29.17	[20.83, 41.67]	[16.67, 55.77]
decision	P	18	33.33	4.17	33.33	[29.17, 37.50]	[16.67, 53.21]
upstream	B	9	90.48	4.12	92.86	[85.71, 92.86]	[85.71, 100.00]
upstream	AS	9	97.62	4.12	100.00	[92.86, 100.00]	[90.48, 100.00]

Three executions per row. SD uses ddof=1. Intervals resample whole blocks and paired execution indices together (10,000 draws, seed 20261102). Finite BA draws: decision 9612/10,000; upstream 6986/10,000. Other undefined counts are in JSON; return intervals use all draws. Conditional finite-draw percentiles are descriptive, with only three runs/blocks; no calibrated-coverage claim. Panel support differs and is not pooled.

Table A.30: Net account-return dispersion on fixed panel cycles (percent).

Panel	Variant	Accounts	Mean	SD	Median	Range	Descriptive interval
decision	B	13	-1.79	0.24	-1.74	[-2.04, -1.57]	[-4.42, -1.24]
decision	A	13	-1.57	0.39	-1.59	[-1.95, -1.17]	[-4.42, -0.70]
decision	S	13	-1.73	0.19	-1.63	[-1.95, -1.61]	[-3.18, -0.83]
decision	AS	13	-1.54	0.68	-1.65	[-2.16, -0.82]	[-4.42, -0.43]
decision	P	13	-1.45	0.34	-1.43	[-1.79, -1.11]	[-4.42, -0.58]
upstream	B	9	0.28	0.14	0.37	[0.12, 0.37]	[-1.15, 2.26]
upstream	AS	9	0.62	0.22	0.75	[0.37, 0.75]	[-0.51, 2.26]

Each return is a mean over 13 decision-panel or nine upstream-panel symbol–block accounts. Accounts use the same frozen prices/costs and only panel cycles; not a portfolio or the broad 51-account result. Three executions per row. SD uses ddof=1. Intervals resample whole blocks and paired execution indices together (10,000 draws, seed 20261102). Finite BA draws: decision 9612/10,000; upstream 6986/10,000. Other undefined counts are in JSON; return intervals use all draws. Conditional finite-draw percentiles are descriptive, with only three runs/blocks; no calibrated-coverage claim. Panel support differs and is not pooled.

Table A.31: Paired balanced-accuracy differences by execution (percentage points).

Panel	Contrast	Mean	SD	Median	Range	Descriptive interval
decision	AS-B	5.56	14.63	4.17	[-8.33, 20.83]	[-9.56, 23.08]
decision	A-B	5.56	9.62	0.00	[0.00, 16.67]	[-6.06, 17.95]
decision	AS-S	2.78	10.49	4.17	[-8.33, 12.50]	[-16.67, 12.50]
decision	AS-A	-0.00	7.22	4.17	[-8.33, 4.17]	[-11.54, 6.25]
decision	S-B	2.78	4.81	0.00	[0.00, 8.33]	[-7.58, 12.50]
decision	AS-P	-2.78	6.36	-4.17	[-8.33, 4.17]	[-19.23, 3.12]
upstream	AS-B	7.14	0.00	7.14	[7.14, 7.14]	[0.00, 14.29]

Three executions per row. SD uses ddof=1. Intervals resample whole blocks and paired execution indices together (10,000 draws, seed 20261102). Finite BA draws: decision 9612/10,000; upstream 6986/10,000. Other undefined counts are in JSON; return intervals use all draws. Conditional finite-draw percentiles are descriptive, with only three runs/blocks; no calibrated-coverage claim. Panel support differs and is not pooled.

Appendix A.3. Factorial Prompt Delivery Audit

The R4 audit publishes the exact panel support, per-prompt section/length/hash inventory, all 474 research-prompt delivery checks, the 18 training donor maps with 90 rank-slot mappings, 54 paired isolation checks, per-run metric and interaction distributions, and the seven-contrast finite-draw envelope. The source archive and unchanged raw requests substantiate the missing factor-value rows; flags or numerical features alone do not establish what the LLM received. The separate repaired bank now has 270 verified fresh decisions. Its completed audit, both-class metrics, equal-symbol macros, cutoff disagreement and 195 panel-account executions with daily ledgers are retained independently of the old compressed-report artifacts.

Table A.32: Existing 18-unit report/consensus contrasts, not full factor-value effects (percentage points).

Contrast	Mean BA	SD	Descriptive interval	Mean LONG
AS-B	5.56	14.63	[-9.56, 23.08]	-12.96
A-B	5.56	9.62	[-6.06, 17.95]	-7.41
AS-S	2.78	10.49	[-16.67, 12.50]	-9.26
S-B	2.78	4.81	[-7.58, 12.50]	-3.70
AS-A	-0.00	7.22	[-11.54, 6.25]	-5.56
AS-P	-2.78	6.36	[-19.23, 3.12]	-1.85
interaction	-2.78	6.36	[-22.92, 12.12]	-1.85

Three paired executions; 10,000 joint block/run draws. BA uses 9,612 finite draws. Interaction is AS-A-S+B, added after inference. Only three blocks/runs; no calibrated coverage or significance claim. Full factor-value rows were absent from all alpha-bearing archived prompts.

Table A.33: Corrected full-value panel: three executions per variant on the same 18 units.

Variant	Acc.	BA	BA SD	MCC	LONG	Return
B	33.33	29.17	0.00	-0.395	44.44	-1.640
A	48.15	40.28	2.41	-0.200	29.63	-1.029
S	29.63	26.39	2.41	-0.446	48.15	-1.820
AS	53.70	44.44	2.41	-0.121	24.07	-0.670
P	44.44	37.50	4.17	-0.249	33.33	-1.121

Percentages except MCC. Each score pools 18 units within one execution, then averages three scores; equal-symbol macros and defined denominators are in JSON. Six UP/twelve DOWN per execution; eight symbols, five dates, three blocks. Return averages 13 panel symbol–block accounts per execution, not a portfolio. CASH references: accuracy 66.67, BA 50.00, return zero; MCC undefined. No pooling with the broad 102-unit compressed-report study.

Table A.34: Corrected paired BA contrasts and the predefined seven-contrast descriptive envelope (percentage points).

Contrast	Mean	SD	Pointwise interval	Joint envelope	LONG delta
AS-B	15.28	2.41	[4.76, 21.57]	[1.79, 28.77]	-20.37
A-B	11.11	2.41	[-2.38, 21.79]	[-2.38, 24.60]	-14.81
AS-S	18.06	4.81	[10.42, 23.53]	[4.56, 31.55]	-24.07
S-B	-2.78	2.41	[-10.42, 3.85]	[-16.27, 10.71]	3.70
AS-A	4.17	4.17	[-5.13, 10.00]	[-9.33, 17.66]	-5.56
AS-P	6.94	4.81	[2.08, 12.82]	[-6.55, 20.44]	-9.26
interaction	6.94	6.36	[-8.97, 16.67]	[-6.55, 20.44]	-9.26

10,000 paired block/run draws, seed 20261102; 9,612 jointly finite and 388 undefined BA draws. Joint max-deviation radius 13.49 points; not calibrated simultaneous confidence or FWER control. AS-P pointwise interval excludes zero but its joint envelope includes zero. Interaction is AS-A-S+B. Only three executions/blocks; no significance or isolated alpha-value conclusion.

Table A.35: Completed actual-prompt and response audit for the repaired R4 panel.

Check	Observed
Fresh decisions / successful responses	270 / 270
Attempts / JSON errors / new vision calls	274 / 4 / 0
Paired groups / order matches	54 / 54
Alpha-bearing successes with five scalar rows	162 / 162
Old-study successful requests copied	0
Panel account executions / accounts per run	195 / 13

All 54 groups share archived upstream text/parents/rules; A/AS alpha matches, S/AS/P sentiment matches, and AS/P differ only in alpha content with five current factor names. Whole training donor vectors retain schema/masks; rank mapping is not named-factor covariance. Raw failures, responses and checkpoints are retained. All source/runtime pins verified; no new model calls during finalization.

Appendix B. Agent System Prompts

This appendix presents the system and user prompts deployed in the agents.

The following examples are English illustrations for readability, not verbatim archives of every inference request. Actual rendered messages are retained in the request archive; the evaluated compact Decision prompts are in Vietnamese. The target-contract audit distinguishes these actual messages from translated and typeset templates.

Appendix B.1. Decision Agent Prompt

The Decision Agent predicts the profitable-cycle net label in Eq. (4): next-bar Open to target Close, including fees and slippage on both legs. The raw `forecast_horizon` string in the archived study is `T+2.5`, a legacy display token used for the three-bar daily horizon; it is not a settlement-feasibility certificate or a calendar-day count. It is retained in the historical requests rather than retroactively renamed. The excerpt below is generated from the unchanged compact English builder with placeholder reports; prose is line-wrapped and the example JSON is pretty-printed for publication. It does not replace the archived Vietnamese messages or claim that additional index equations were supplied to the model.

```
You are a Vietnamese-market quant trader. Asset: {stock_name}; timeframe:
{time_frame}; target: {h_desc} ({h_val}).

CONTRACT: Choose only LONG or SHORT. LONG only when buying with all cash at
entry Open and selling at target Close remains profitable after the 0.25%
broker fee and 0.10% slippage on EACH leg (~0.70% round trip). Otherwise
choose SHORT, keep CASH for the horizon, never short-sell, and never carry
shares into the next test. Rule: {h_note}

5 REPORTS:
### [1] TREND ANALYSIS
{trend_report}

### [2] CANDLESTICK PATTERNS
{pattern_report}

### [3] TECHNICAL INDICATORS
{indicator_report}

### [4] QUANTITATIVE ALPHA FACTORS
{alpha_report}

### [5] NEWS & MARKET SENTIMENT
{sentiment_report}

EVIDENCE PRIORITY:
1. Trend/Pattern define regime, barriers and timing; Indicator confirms
momentum.
2. Normalized Alpha is quantitative evidence: it may veto a flow-less
```

```
breakout or lead a reversal only with >=3/5 agreement.
3. Sentiment has the lowest weight and never overrides price or Alpha.
4. Conflict gate: when Trend and Pattern both confirm a downtrend, do not
choose LONG merely from 1-2 positive alphas or upbeat news. Counter-trend
LONG requires major support, dominant reversal Alpha, and candle/indicator
confirmation; otherwise choose SHORT.

Return one valid JSON object only; no Markdown or text outside JSON:
{
  "decision": "LONG or SHORT",
  "forecast_horizon": "{h_val}",
  "confidence": "Very high|High|Medium|Low",
  "risk_reward_ratio": 1.0,
  "evidence_for": "max 2 sentences",
  "evidence_against": "max 1 sentence",
  "justification": "max 2 sentences"
}
```

Full receives five report sections, Base three, and Alpha-only or Sentiment-only four. Alpha and sentiment sections are independently included under their respective variant flags. The fixed prompt elicits the primary net target with 0.25% fees and 0.10% slippage per leg. Other-cost execution sensitivities preserve these predictions and labels; they are not newly elicited classifiers. New LLM-backed runs reject a different label-cost configuration until a separate prompt/protocol is versioned.

Appendix B.2. Pattern Agent Prompt

The frozen configuration names the vision-language model `qwen/qwen3.8-27b`; the following Pattern Agent prompt is prototype source, not a provenance-validated research artifact:

```
/no_think
You are an AI specializing in Price Action pattern analysis.

ABSOLUTE RULES:
1. ONLY return the 6 fields below. Do NOT add any other fields.
2. NO reasoning, NO explanations.
3. NO fallback scenarios. Provide exactly ONE definitive assessment.
4. Each field on ONE line. Be concise.

MANDATORY FORMAT:
**Pattern:** <Name>
**Confidence:** High | Medium | Low
**Forecast Bias:** Bullish | Bearish | Neutral
**Evidence:** <Brief description>
**Key Candle:** <List key candle>
**Trade Implication:** <Scenario>
```

The 16 reference patterns provided to the model include: Inverse Head and Shoulders, Double Bottom, Falling/Rising Wedge, Triangle, Bull/Bear Flag, Rectangle, V-Reversal, and Clear Trend.

If the vision model fails, the agent falls back to a text-only analysis of the most recent 20 OHLCV rows using the text LLM (`openai/gpt-oss-20b`).

Appendix B.3. Trend Agent Prompt

The Trend Agent also utilizes a vision-language model to evaluate dynamically generated support and resistance lines.

```
You are an expert at analyzing K-line trend patterns in high-frequency trading.
Your sole objective: PREDICT {h_desc}.
You must always reply in the exact markdown format requested.
NEVER write all fields on a single line.

This is a {time_frame} candlestick chart with automatic trendlines:
- **Blue line** = support line (derived from close prices)
- **Red line** = resistance line (derived from close prices)

Your tasks:
1. Analyze how price interacts with the blue and red lines.
2. Determine if candles are bouncing, breaking out, or compressing.
3. Evaluate the slope and distance between the two lines.
4. Predict the direction for **{h_val}**: bullish, bearish, or sideways.

MANDATORY FORMAT:
**Trend Direction:** Bullish | Bearish | Sideways
**Support Level:** <specific price>
**Resistance Level:** <specific price>
**Trendline Slope:** Rising | Falling | Flat
**Price vs Support:** <bouncing / breaking / compressing>
**Detailed Analysis:** <2-3 sentences of price action analysis>
**Trend Prediction:** <1-2 sentences forecasting {h_desc}>
**Confidence:** High | Medium | Low
```

Appendix B.4. Indicator Agent Prompt

The Indicator Agent receives a pre-calculated table of technical indicators and their deterministic classifications (computed in Python). Its prompt forces it to adhere strictly to the system's hardcoded signals rather than re-interpreting the raw data.

```
You are an expert at High-Frequency Trading technical analysis.
SOLE OBJECTIVE: Predict price direction for **{h_desc}**.

MOST IMPORTANT RULES:
1. SIGNAL CLASSIFICATION and CONSENSUS have already been determined
```

```
   by the system - you MUST use these exact conclusions. Do NOT count
   or change them.
2. Each indicator has a strict [END ... SECTION] boundary. Do NOT use
   numbers from one section to describe another.
3. Your only task: write a clear, actionable qualitative interpretation
   for {h_short}.
4. NO hallucinated numbers. NO saying 'missing information'.

## Data Table (Calculated & Classified)
[Pre-rendered Table and System Consensus injected here]

## Requirement: Write an interpretation report
For EACH indicator, use this exact template:
**[Indicator Name]**
- {h_short} Signal: [EXACT system classification]
- Key levels/crossovers: [Specific numbers from the correct section]
- Momentum: [accelerating / decelerating / stable] - short explanation

Then summarize the final consensus:
**Signal Convergence Summary - Prediction for {h_short}**
- Bullish agreement: **{n_tang}/{total}**
- Bearish agreement: **{n_giam}/{total}**
- Primary Trend: **{consensus}** - Confidence: **{confidence}**
- Conclusion: [1-2 sentences summarizing potential movement]
```

Appendix B.5. Sentiment LLM Prompt (Subroutine of Alpha Agent)

Sentiment Analysis is invoked programmatically as a subroutine within the Alpha Agent node rather than acting as a standalone agent in the DAG. ViSoBERT runs through local Transformers by default and can instead use a configured Hugging Face Dedicated Inference Endpoint; a deterministic lexicon is the explicit last-resort fallback. The actual scorer, backend, and fallback flag are retained per newly scored article; the text LLM only synthesizes a qualitative summary of the resulting news bias and related-company impacts.

```
You are an expert at analyzing sentiment in the VN stock market.
Synthesize data from press articles. Reply in markdown.

Analyze sentiment for **{target}** ({time_frame}).

=== AGGREGATE RESULTS ===
Article Count: {article_count} | Avg Score: {avg_score} | Label: {label}
Distribution: {positive} / {neutral_count} / {negative}
Sentiment Bar: {sentiment_bar_visual} | Model: {model_note}

=== LATEST 15 ARTICLES ===
{art_sum}
```

```
=== RELATED COMPANIES ===
{rel_sum}

---
Write the report using this template:

**Sentiment Summary - {target}**
**Main Takeaway:** [1-2 sentences]
**Detailed Analysis:**
- News Trend: [...]
- Positive Signals: [...]
- Negative Signals: [...]
- Pressure from Related Companies: [...]
```

Appendix B.6. Alpha Agent LLM Reasoning Prompt

In live/prototype mode the Alpha Agent may invoke the text LLM after computing quantitative factors; the backtest uses deterministic factor evidence. The following legacy prompt illustration retains the words "order flow as source wording. Its inputs are OHLCV proxies, not measured signed trades or order-book flow:

```
You are a quantitative and order flow analyst forecasting {horizon_label}.
Below are OHLCV-only alpha factors and a separate sentiment context
for **{stock_name}**.

### ALPHA FACTORS RESULTS
{report_md}

### SEPARATE SENTIMENT & NEWS CONTEXT
{sentiment_context_md}

FUSION CONTRACT:
- Alpha values and ranks use OHLCV only.
- Sentiment does not enter factor formulas; it is text context only.

Compare the two evidence sources qualitatively to:
1. Provide a short comment (1-2 sentences) on current momentum.
2. Synthesize: Are flow and sentiment in agreement or contradiction?
3. Identify hidden risks/opportunities from news not captured by technical alphas.

Do NOT over-analyze. Deduce from the numbers to conclude the view for {horizon_label}.
```

Appendix C. LangGraph State Schema

The pipeline uses `IndicatorAgentState`, a Python `TypedDict`. Table C.36 lists selected runtime state fields; it is not a complete declaration or a list of realised evaluation outcomes. The backtest wrapper keeps post-inference target metadata separately, including boundary e, decision $d = e - 1$, entry e, and target $e+L-1$. Observed-bar positions and dates are distinct; no calendar-day addition is used to manufacture a bar.

Table C.36: Selected runtime state fields (`IndicatorAgentState`)

Field	Type	Description
Input fields (set before graph execution)		
`kline_data`	`dict`	OHLCV dictionary for technical indicator computation
`as_of_date`	`timestamp`	Final observable decision-bar timestamp; no entry or target data
`time_frame`	`str`	Time period for K-line data (e.g., "1 ngày")
`stock_name`	`str`	Stock ticker symbol
`is_backtest`	`bool`	Flag to block external crawlers in backtest mode
`pattern_image`	`str`	Base64-encoded candlestick chart PNG
`trend_image`	`str`	Base64-encoded trendline-annotated chart PNG
Declared indicator arrays (not populated by this node)		
`rsi`	`List[float]`	RSI values (period 14)
`macd, macd_signal, macd_hist`	`List[float]`	MACD line, signal, histogram (12, 26, 9)
`stoch_k, stoch_d`	`List[float]`	Stochastic %K, %D (14, 3, 3)
`roc`	`List[float]`	Rate of Change (period 10)
`willr`	`List[float]`	Williams %R (period 14)
`bb_upper, bb_middle, bb_lower`	`List[float]`	Bollinger Bands (20, 2)
`adx, plus_di, minus_di`	`List[float]`	ADX and directional indicators
Actual Indicator Agent output		
`indicator_report`	`str`	Report with five-indicator consensus
Alpha Agent outputs		
`alpha_report`	`str`	Five alpha factors + LLM reasoning
Pattern Agent outputs		
`pattern_report`	`str`	Six-field markdown pattern analysis
Trend Agent outputs		
`trend_report`	`str`	Eight-field markdown trend analysis
Sentiment outputs (written by Alpha Agent internally)		
`sentiment_report`	`str`	CafeF sentiment summary
`sentiment_data`	`Dict[str, Any]`	Display articles plus complete compact cutoff-safe observations
`sentiment_norm`	`Dict[str, Any]`	Time-decayed sentiment, relative context, availability and provenance
Decision Agent outputs		
`final_trade_decision`	`str`	Normalised parsed decision JSON; raw responses archived separately
`decision_prompt`	`str`	Full prompt sent to the Decision Agent
`messages`	`List[BaseMessage]`	LangChain message history

Evaluation records and historical aliases.. Table C.37 documents the canonical outcome fields stored outside graph inputs. New engine, robustness and ablation records derive these fields from the exact evaluation frame. The fields are unavailable to the model at decision time. Net and gross returns are distinct fractions; legacy percent fields remain for compatibility. A zero return maps to DOWN in both targets; SHORT is a DOWN forecast and CASH execution in the long-only ledger. Historical `actual_prev_close` is not a trustworthy universal alias for `decision_close`: 178 of 600 retained legacy point records equal `entry_open`, whereas the others differ. Equality alone does not establish

the field's historical meaning. Without authenticated original snapshots, the canonical decision close remains unavailable for those records; no legacy label, price or prediction is rewritten to fit the explanation.

Table C.37: Canonical evaluation fields recorded after inference, outside graph inputs.

Field	Meaning
`decision_close`	Observed C_{e-1}, never substituted for entry.
`entry_open`	Candidate next-bar entry O_e.
`exit_close`	Immutable target close C_{e+L-1}.
`gross_return`	Fraction $C_{e+L-1}/O_e - 1$, before costs.
`net_return`	Candidate LONG fraction after both cost legs, Eq. (4); not necessarily an executed account return.
`label_net`	Primary UP iff net return > 0; DOWN otherwise.
`label_gross`	Secondary UP iff gross return > 0; DOWN otherwise.
`decision_bar_index`	Zero-based final observed bar $e - 1$.
`target_horizon_bars`	Complete observed-bar horizon L, not calendar days.
`label_fee_each_side`	Fee rate used for the immutable label.
`label_slippage_each_side`	Slippage rate used for the immutable label.
`label_contract_version`	`net_cycle_v1` for new records; `legacy_unverified` when canonical provenance is missing.

Canonical returns are fractions; legacy `actual_pct_change` is net percent and `execution_pct_change` is gross percent. New records retain `actual_direction` as a net-label alias. Historical `actual_prev_close` is not automatically renamed: its meaning is unverified without the original snapshot. SHORT is a DOWN forecast with CASH execution, not a short sale. Zero returns map to DOWN; incomplete horizons are not shortened. Execution stress costs/delays do not relabel the fixed target.

Graph topology.. The DAG is constructed in `graph_setup.py` using the LangGraph `StateGraph` API. In the Full System configuration, the sequential edge order is:

`START` → `Indicator` → `Alpha` → `Pattern` → `Trend` → `Decision` → `END`

During evaluation, graph execution is split into a variant-independent upstream graph and variant-specific decision graphs. Indicator, Pattern, and Trend run once per exact cutoff; the resulting state is deep-copied into Full, Alpha-only, Sentiment-only, and Baseline branches as required by the experiment. This shared-artifact topology prevents Alpha settings from changing the No-Alpha evidence snapshot.

Appendix D. Custom Alpha Formula Implementation

This appendix shows the computational cores of the five custom alpha formulas listed in Table 2; the snippets omit metadata and are not complete function declarations. All formulas operate on the technical variables extracted from the OHLCV window by the function `_extract_tech_vars()`.

Appendix D.1. α_1: Flow-Driven Momentum (FDM)

```
def _alpha1_fdm(tv: dict) -> dict:
    roc1_adj  = tv["roc1_adjusted"]    # ROC(1) / rolling_std(ROC, 20)
    vol_surge = tv["vol_surge"]        # Volume / ADV(20)
    raw   = roc1_adj * vol_surge * 0.5
    value = math.tanh(raw)
    # Signal threshold: |value| > 0.10
```

Where `roc1_adjusted` is the one-period rate of change normalised by its 20-period rolling standard deviation, and `vol_surge` is the ratio of current volume (or intra-bar range proxy when volume is unavailable) to the 20-period average volume.

Appendix D.2. α_2: MACD-Flow Proxy (MFP)

```
def _alpha2_mfp(tv: dict) -> dict:
    roc1_adj = tv["roc1_adjusted"]
    macd_hist = tv["macd_hist_cur"]
    z_macd = tanh(macd_hist / (close * 0.005)) * 1.5
    raw = z_macd + roc1_adj * 0.5
    value = math.tanh(raw)
```

This factor always uses price-derived MACD histogram and ROC inputs. Sentiment coverage does not switch or modify its formula; sentiment is provided only to the later LLM text-context comparison.

Appendix D.3. α_3: Liquidity Void Reversion (LVR)

```
def _alpha3_lvr(tv: dict) -> dict:
    z_close_dev5 = tv["zscore_close_dev5"]
    raw   = -1.0 * z_close_dev5 * 0.8
    value = math.tanh(raw)
    # Signal threshold: |value| > 0.15 (higher than other alphas)
```

Where `zscore_close_dev5` is the z-score of the series $\frac{C_u - \mathrm{SMA}_5}{C_u}$ over a 20-period rolling window at bar position u. The negative sign creates a mean-reversion signal: when the close is abnormally far above SMA(5), the alpha predicts a pullback.

Appendix D.4. α_4: Bollinger Squeeze & Flow (BFE)

```
def _alpha4_bfe(tv: dict) -> dict:
    bb_pct_b = tv["bb_pct_b"]         # Bollinger %B
    vol_z5   = tv["vol_zscore_5"]     # Volume z-score (5-period)
    raw   = (bb_pct_b - 0.5) * vol_z5 * 1.5
    value = math.tanh(raw)
```

This is a signed interaction, not a positive-volume confirmation gate. For $\%B > 0.5$, positive VolZ_5 yields a positive value and negative VolZ_5 yields a negative value; for $\%B < 0.5$ these directions reverse. Either $\%B = 0.5$ or $\mathrm{VolZ}_5 = 0$ gives zero. Nonzero absolute volume z-score only means a deviation from its reference mean, not expansion. Neither band width nor a preceding crossing is tested. The prototype classifies the unrounded value as BULLISH if $v > 0.10$, BEARISH if $v < -0.10$, and NEUTRAL otherwise, then displays four decimals.

Table D.38: Synthetic signed-volume examples for the unchanged custom α_4.

$\%B$	VolZ_5	Raw product	tanh(raw)	Signal
0.25	-1	+0.375	+0.3584	BULLISH
0.25	+0	-0.000	-0.0000	NEUTRAL
0.25	+1	-0.375	-0.3584	BEARISH
0.50	-1	-0.000	-0.0000	NEUTRAL
0.50	+0	+0.000	+0.0000	NEUTRAL
0.50	+1	+0.000	+0.0000	NEUTRAL
0.75	-1	-0.375	-0.3584	BEARISH
0.75	+0	+0.000	+0.0000	NEUTRAL
0.75	+1	+0.375	+0.3584	BULLISH

Arithmetic fixtures, not market observations or performance estimates. Signals use the prototype's strict ±0.10 thresholds before value rounding. Negative volume z-scores reverse sign; zero volume z-score or $\%B = 0.5$ suppresses the interaction. No band-width squeeze gate is present.

Both the prototype and registry retain this algebraic interaction. Their preprocessing differs: the prototype uses its own population-standard-deviation helpers and rounded extracted features, whereas the rolling registry uses sample standard deviations. The synthetic registry oracle checks that implementation separately; identical general preprocessing or all reported factor values is not claimed. Legacy breakout/squeeze wording in metadata or prompts does not override the signed formula. No formula or archived prediction has been changed.

Appendix D.5. α_5: Order Flow Exhaustion (OFE)

```
def _alpha5_ofe(tv: dict) -> dict:
    roc1_adj  = tv["roc1_adjusted"]
    close_pos = tv["close_pos"]   # (Close - Low)/(High - Low) - 0.5
    raw   = roc1_adj * close_pos * 3.0
    value = math.tanh(raw)
```

Where `close_pos` $\in [-0.5, 0.5]$ measures where the closing price falls within the intra-bar range, centred at zero. A positive `close_pos` with positive momentum produces a positive price-action proxy; it does not establish measured buying pressure; the $\times 3.0$ amplification increases signal responsiveness.

Appendix E. Indicator Signal Classification Logic

The following rules describe `_classify_signals` in execution order. Lists are independently filtered to finite values and rounded to four decimals before classification. Subscripts below refer to the last retained values, not necessarily adjacent bars when inputs have missing values. Missing/error indicators vote NEUTRAL and remain in the five-vote denominator; this fallback is not evidence of neutral market conditions. The live prompt requires the supplied signals/counts to be copied, while the backtest uses the deterministic report directly.

Appendix E.1. MACD Classification

First test $m_{-1} < s_{-1}$ and $m_0 \geq s_0$ for BULLISH, then $m_{-1} > s_{-1}$ and $m_0 \leq s_0$ for BEARISH. Equality on the previous value is not a crossover. Without a cross, $h_0 > 0$ is BULLISH only if $h_0 \geq h_{-1}$ (or no previous histogram exists); $h_0 < 0$ is BEARISH only if $h_0 \leq h_{-1}$. Contraction or zero is NEUTRAL. If fewer than two MACD/signal values exist, a finite histogram's sign alone is used. Missing histogram without a cross yields NEUTRAL.

Appendix E.2. RSI Classification

Test in order: $r_0 \geq 70$ gives BEARISH; $r_0 \leq 30$ gives BULLISH; $55 \leq r_0 < 70$ gives BULLISH; $30 < r_0 \leq 45$ gives BEARISH. In $45 < r_0 < 55$, at least three retained values are required: compare r_0 with r_{-2} for BULLISH/BEARISH when greater/less; equality or insufficient history gives NEUTRAL. The earlier "otherwise NEUTRAL" description omitted this trend fallback.

Appendix E.3. Stochastic Oscillator Classification

With two K/D values, test $K_{-1} < D_{-1}$ and $K_0 \geq D_0$ for BULLISH, then $K_{-1} > D_{-1}$ and $K_0 \leq D_0$ for BEARISH. Crossovers override extreme zones. Otherwise test $K_0 \geq 80$ for BEARISH, then $K_0 \leq 20$ for BULLISH. In the middle zone, the current implementation returns BULLISH only when both current K/D are truthy and $K_0 > D_0$; all other middle-zone cases, including equality or $D_0 = 0$, return BEARISH. This is a documented legacy fallback, not a recommended zero-handling convention. With fewer than two K/D values, only extreme-zone rules apply and the middle zone is NEUTRAL; absent K is NEUTRAL.

Appendix E.4. Williams %R Classification

First $w_0 \geq -20$ gives BEARISH, then $w_0 \leq -80$ gives BULLISH. In $-80 < w_0 < -20$, compare w_0 and w_{-2} when at least three retained values exist: greater gives BULLISH, less BEARISH; equality or insufficient history gives NEUTRAL. Extreme thresholds override this trend rule.

Appendix E.5. ROC Classification

ROC is in percentage units: $r_0 > +0.5\%$ gives BULLISH; $r_0 < -0.5\%$ gives BEARISH. Equality at either boundary, missing values and the interior interval give NEUTRAL. Four-decimal cleaning precedes the comparison.

Appendix E.6. Consensus Rule

All five indicators contribute to the count, including missing/error NEUTRAL votes. More bullish than bearish votes gives BULLISH; more bearish gives BEARISH; equal counts give MIXED. A directional count of at least four gives High confidence, three Medium, and a smaller directional plurality Low. A tie always has Low confidence. These are deterministic labels, not calibrated probabilities. The release provides 67 indicator boundary/fallback cases, five consensus cases, eleven prototype Alpha_4 cases and three independent registry oracles; these synthetic cases are not market-performance observations.

Appendix F. Software Dependencies and Reproducibility

Table F.39 lists the core software dependencies used in the implementation. All packages are specified in `requirements.txt` and are installable via `pip`.

Sentiment scorer provenance.. The runtime no longer sends this model to the deprecated generic Hugging Face serverless route. With the default automatic backend policy, it calls the exact Dedicated Endpoint URL supplied through the environment only when one is configured; otherwise it loads `5CD-AI/Vietnamese-Sentiment-visobert` locally. A failed dedicated endpoint in automatic mode also falls back to this local path. Local inference loads the Transformers sequence-classification model and the repository's fast `tokenizer.json`, rather than the incompatible legacy slow XLM-R tokenizer declaration, and applies truncation at 256 tokens. The official class mapping is `LABEL_0/NEG`

Table F.39: Software dependencies

Package	Version	Purpose
`flask`	latest	Web interface for real-time demo
`pandas`	latest	Data manipulation and OHLCV processing
`numpy`	latest	Numerical computation
`matplotlib`	latest	Chart rendering backend
`mplfinance`	latest	Candlestick chart generation
`scipy`	latest	Statistical functions
`TA-Lib`	latest	Technical indicator computation (MACD, RSI, etc.)
`langchain`	latest	LLM abstraction layer
`langchain-groq`	latest	Groq API integration for LLM inference
`langgraph`	latest	Multi-agent DAG orchestration framework
`vnstock`	$\geq 3.0.0$	Vietnamese stock market data (VCI/MSN sources)
`requests`	latest	HTTP client for CafeF crawling and optional Dedicated Endpoint inference
`transformers`	latest	Local ViSoBERT model and fast-tokenizer pipeline
`torch`	latest	Local CPU/CUDA tensor inference
`sentencepiece`	latest	ViSoBERT tokenizer compatibility support
`beautifulsoup4`	latest	HTML parsing for article content extraction
`Pillow`	latest	Image processing utilities

(negative), `LABEL_1/POS` (positive), and `LABEL_2/NEU` (neutral). Only a ViSoBERT failure causes the deterministic lexicon fallback. New records store `scorer`, `scorer_backend`, and `is_fallback`.

The supplied historical cache schema stores label, numeric score, and confidence but predates those provenance fields. The audit therefore does not infer a scorer from numerical values: every such record is classified as `legacy-unknown`, and it contributes to a verified ViSoBERT or lexicon count only when explicit metadata is present. This conservative rule prevents a runtime configuration from being treated as evidence about an older article.

Web result serialization.. Background analysis can return NumPy or Pandas scalars even after all six agent steps have completed. Before Flask status endpoints call `jsonify`, the result payload is now normalized recursively to native JSON-compatible Python types. NumPy Boolean, integer, and floating scalars are converted to their Python counterparts; arrays and tabular vectors become lists; timestamps use ISO 8601; and non-finite floating values become JSON `null`. This transport-layer normalization prevents a completed job from returning HTTP 500 solely because a nested `numpy.bool_` or related object is not serializable. It does not alter the decision, sentiment score, or archived experimental metric.

Reproducibility artifacts.. The following artifacts are provided or specified for reproducibility:

- **Model versions**: The reported text default is `openai/gpt-oss-20b`; the vision default is `qwen/qwen3.8-27b`. Both use the Groq API. The backtest engine resolves these configurations directly and stores the effective run settings with its result artifacts.

Table F.40: Ticker-selection strata and point-in-time sentiment coverage.

Ticker	Sector	Design stratum	Dated/total	V/L/U	Mean 90d	≥ 8 points
BHN	Food/beverages	State-associated, specialised	20/20	20/0/0	1.15	0/20
CMG	Information technology	Mid-cap	20/20	20/0/0	0.85	0/20
FPT	Information technology	VN30/large-cap	20/20	20/0/0	1.45	2/20
HVN	Aviation and tourism	State-associated	20/20	20/0/0	9.45	19/20
MBB	Banking	VN30/large-cap	20/20	20/0/0	7.35	7/20
MWG	Retail	VN30/large-cap	20/20	20/0/0	6.05	8/20
VCB	Banking	VN30/large-cap	20/20	20/0/0	9.90	18/20
VJC	Aviation and tourism	Large private carrier	20/20	20/0/0	10.10	17/20
VNM	Consumer food	VN30/large-cap	20/20	20/0/0	5.60	2/20
All	**Nine-symbol purposive sample**		**180/180**	**180/0/0**	**5.77**	**73/180**

Notes: V/L/U denotes verified ViSoBERT, verified lexicon, and unknown scorer counts. "Mean 90d" is the mean number of dated articles in the inclusive 90-day lookback at the 20 test cutoffs. The final column counts test points with at least eight eligible articles. The underlying machine-readable audit is generated by `scripts/analyze_sentiment_coverage.py`.

- **Temperature**: The declared default is $T = 0.0$. In the new archived study, the installed client coerces zero to 10^{-8}, which is the value recorded in all 756 attempt metadata records (Section 5.8). These client metadata are not a wire capture or authentication of served weights, and do not imply deterministic output. Legacy runs without these records remain unverified.
- **Retry policy**: The application policy defaults to at most three text calls (two retries with 5/10-second exponential waits) and two vision calls (one retry with a 3-second wait) when SDK retries are disabled. Positive SDK `max_retries` budgets make the SDK the single retry owner and reduce the application wrapper to one call; SDK-internal attempts are not observable and remain unknown in telemetry. Inter-test pacing is recorded separately from retry delays.
- **Historical sentiment cache**: Nine JSON files (`sentiment_cache_{SYMBOL}.json`), one per evaluated stock, contain pre-crawled and scored CafeF articles. Point-in-time filtering uses the parsed publication date; undated records are rejected in strict research mode. The supplied files lack scorer metadata, so all 180 cached scores are labelled `legacy-unknown` in the coverage audit.
- **Default configuration**: Stored in `default_config.py` with all hyperparameters.

References

[1] F. Xiong, X. Zhang, A. Feng, S. Sun, C. You, Quantagent: Price-driven multi-agent llms for high-frequency trading, arXiv preprint arXiv:2509.09995 (2025).
URL `https://arxiv.org/abs/2509.09995`

[2] LangChain AI, Langgraph documentation, `https://www.langchain.com/langgraph` (2024).

[3] H. Yang, X.-Y. Liu, C. D. Wang, et al., Fingpt: Open-source financial large language models, arXiv preprint arXiv:2306.06031 (2023).
URL `https://arxiv.org/abs/2306.06031`

[4] S. Wu, O. Irsoy, S. Lu, et al., Bloomberggpt: A large language model for finance, arXiv preprint arXiv:2303.17564 (2023).
URL https://arxiv.org/abs/2303.17564

[5] T. Research, Tradingagents: Multi-agents llm financial trading framework, arXiv preprint arXiv:2412.20138 (2024).
URL https://arxiv.org/abs/2412.20138

[6] A. Tsantekidis, N. Passalis, A. Tefas, et al., Forecasting stock prices from the limit order book using convolutional neural networks, IEEE 19th Conference on Business Informatics (2017) 7–12doi:10.1109/CBI.2017.23.

[7] Y. Deng, F. Bao, Y. Kong, Z. Ren, Q. Dai, Deep direct reinforcement learning for financial signal representation and trading, IEEE Transactions on Neural Networks and Learning Systems 28 (3) (2016) 653–664. doi:10.1109/TNNLS.2016.2522401.

[8] A. W. Lo, H. Mamaysky, J. Wang, Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation, The Journal of Finance 55 (4) (2000) 1705–1765. doi:10.1111/0022-1082.00265.

[9] Y. Han, C. Zhang, X. Chen, X. Yang, Z. Wang, G. Yu, B. Fu, H. Zhang, Chartllama: A multimodal llm for chart understanding and generation, arXiv preprint arXiv:2311.16483 (2023).
URL https://arxiv.org/abs/2311.16483

[10] D. Shu, H. Yuan, Y. Wang, Y. Liu, H. Zhang, M. Du, Finchart-bench: Benchmarking financial chart comprehension in vision-language models, in: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, 2026, pp. 13447–13466. doi:10.18653/v1/2026.acl-long.615.
URL https://aclanthology.org/2026.acl-long.615/

[11] Z. Wang, Martingale doppelgänger-eval: An identification framework for auditing candlestick understanding in vision-language models, arXiv preprint arXiv:2606.17423 (2026). arXiv:2606.17423.
URL https://arxiv.org/abs/2606.17423

[12] N. Jegadeesh, S. Titman, Returns to buying winners and selling losers: Implications for stock market efficiency, The Journal of Finance 48 (1) (1993) 65–91. doi:10.1111/j.1540-6261.1993.tb04702.x.

[13] E. F. Fama, K. R. French, The cross-section of expected stock returns, The Journal of Finance 47 (2) (1992) 427–465. doi:10.1111/j.1540-6261.1992.tb04398.x.

[14] J. Kang, S. Kim, B. Lee, Single-stock alpha factors based on price-volume features, Expert Systems with Applications 168 (2021) 114236. doi:10.1016/j.eswa.2020.114236.

[15] C. R. Harvey, Y. Liu, H. Zhu, ...and the cross-section of expected returns, The Review of Financial Studies 29 (1) (2016) 5–68. doi:10.1093/rfs/hhv059.

[16] B. Hurst, Y. H. Ooi, L. H. Pedersen, A century of evidence on trend-following investing, The Journal of Portfolio Management 44 (1) (2017) 15–29. doi:10.3905/jpm.2017.44.1.015.

[17] S. Varma, R. Simon, Bias in error estimation when using cross-validation for model selection, BMC Bioinformatics 7 (2006) 91. doi:10.1186/1471-2105-7-91.

[18] D. H. Bailey, M. López de Prado, The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality, The Journal of Portfolio Management 40 (5) (2014) 94–107. doi:10.3905/jpm.2014.40.5.094.

[19] P. C. Tetlock, Giving content to investor sentiment: The role of media in the stock market, The Journal of Finance 62 (3) (2007) 1139–1168. doi:10.1111/j.1540-6261.2007.01232.x.

[20] D. Araci, Finbert: Financial sentiment analysis with pre-trained language models, arXiv preprint arXiv:1908.10063 (2019).
URL https://arxiv.org/abs/1908.10063

[21] X. V. Vo, Market efficiency in vietnam's stock market: Evidence from a weak-form test, Journal of Risk and Financial Management 12 (2) (2019) 100. doi:10.3390/jrfm12020100.

[22] T. H. V. Pham, D. T. Nguyen, Herding behavior in emerging equity markets: Evidence from vietnam, The Journal of Asian Finance, Economics and Business 8 (3) (2021) 45–53. doi:10.13106/jafeb.2021.vol8.no3.0045.

[23] H. T. Truong, M. H. Nguyen, Technical trading rules in the vietnamese stock market: Empirical evidence under price limits, Investment Management and Financial Innovations 17 (4) (2020) 12–24. doi:10.21511/imfi.17(4).2020.02.

[24] S. Fatemi, Y. Hu, Finvision: A multi-agent framework for stock market prediction, arXiv preprint arXiv:2411.08899 (2024).

[25] Y. Xia, P. You, T. Wang, F. Liu, H. Qi, X. Wu, S. Zhang, Agentic trading: When llm agents meet financial markets, arXiv preprint arXiv:2605.19337 (2026).

[26] J. Zhang, G. Liu, O. Johansson, H. Yitayew, K. Ohly, G. Li, Prediction arena: Benchmarking ai models on real-world prediction markets, arXiv preprint arXiv:2604.07355 (2026).

[27] A. Arora, R. Malpani, Predictionmarketbench: A swe-bench-style framework for backtesting trading agents on prediction markets, arXiv preprint arXiv:2602.00133 (2026).

[28] Z. Zeng, J. Liu, S. Chen, T. He, Y. Liao, J. Wang, Z. Wang, Y. Yang, L. Yin, M. Yin, Z. Zhu, T. Cai, Z. Chen, J. Chen, Y. Du, X. Gao, J. Guo, L. Hu, J. Jiao, X. Li, J. Liu, S. Ni, Z. Wen, G. Zhang, K. Zhang, X. Zhou, J. Blanchet, X. Qiu, M. Wang, W. Huang, Futurex: An advanced live benchmark for llm agents in future prediction (2025). arXiv:2508.11987.
URL https://arxiv.org/abs/2508.11987

[29] R. C. Grinold, R. N. Kahn, Active Portfolio Management, McGraw-Hill, 1999.

[30] W. F. Sharpe, Mutual fund performance, Journal of Business 39 (1) (1966) 119–138.

[31] J. Wei, et al., Chain-of-thought prompting elicits reasoning in large language models, NeurIPS (2022).

[32] R. Pardo, The Evaluation and Optimization of Trading Strategies, Wiley, 2008.

[33] G. Appel, Technical Analysis: Power Tools for Active Investors, Financial Times Press, 2005.

[34] J. W. Wilder, New Concepts in Technical Trading Systems, Trend Research, 1978.

[35] G. Lane, Lane's stochastic oscillator (1984).

[36] L. Williams, Trading stocks and commodities with the insiders (1973).

[37] J. J. Murphy, Technical Analysis of the Financial Markets, New York Institute of Finance, 1999.